

ROBOWHALES



FRCNC

TEAM 9032 THE ROBOWHALES

RALEIGH, CARY, DURHAM, NC

OVERVIEW

FRCNC is a project that the **RoboWhales** took on to help our team be more competitive.

There's a lot of meaning packed into the name FRCNC,

FRCNC: It's a CNC machine! A CNC (combined with CAD and CAM) can help your team get to the next level in a big way.

FRCNC: At its core, this project is designed to help FRC teams. If your team can build a moderately complex FRC robot, then FRCNC should be well within your grasp.

FRCNC: FRCNC is designed to be open and CONFIGURABLE. You don't need to cut a 24"x48" sheet of polycarbonate in one go? No problem! Copy our CAD and change the sizes to suit your needs. Only cutting wood? Sure! Tweak the design to use a smaller spindle.

FRCNC: We're very proud of the fact that it was designed by students from North Carolina! We have aspirations that our design (and lessons learned) will help teams from around the world get to the next level.



SPECIFICATIONS

2.2KW spindle, 24000 RPM

X, Y, Z dimensions.

Our CNC is cut up into 3 distinct subcomponents (which can all be assembled separately), which are our axes. Each axis is a reflection of a different degree of motion on our CNC (X = travel along bed, Y = travel along gantry, Z = Travel up/down).

X-Axis

- 1225mm of travel (48.22 inches)

Y-Axis

- 750mm of travel (29.53 inches)

Z-Axis

- 150mm of travel (5.91 inches)
- 2.2KW spindle, 24000 RPM
- 0.012" Deflection under 44lb of force (excellent for cutting aluminum)



PARTS LIST

Parts List & Bill of Materials

Here is the link to the BOM along with a parts list:

<https://docs.google.com/spreadsheets/d/1sZHDCqBL8Id-gWcgBUNVI-J4LUe8LgoLk5cTmV6R138/edit?gid=0#gid=0>

Almost all of the parts (screws, nuts, plates) for FRCNC are be metric. Metric machine parts tend to be less expensive than comparable SAE sized (inch) parts.

The 2.2kW spindle works nicely for us, but it may have been overkill. the 1.5kW spindle would probably work fine. Also, the 400W spindle that we used for V1 served us pretty well, but it was limited to 12000 RPM, the 24000 rpm spindle handles aluminum much better.

We initially cut most of our aluminum plates (omniplates, y-to-z axis plate, etc.) out of 1/8" or 1/4" aluminum on an anemic CNC machine, For our V2 (upgraded) machine we uploaded the STEP files for the gantry plates to a service like Fabworks, Xometry, SendCutSend and had them laser cut from 1/4" steel, which was surprisingly cheaper than aluminum (and better in every way because it is stiffer). Teams should weigh their options when deciding whether to cut parts in-house or sent them off. A great example of this is the 6x2 tube we use now. That probably would have cost a fortune to send out, but we did it for less than \$100 because we bought the tube and machined it ourselves on the V1 version of our machine (albeit slowly).



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Tools you'll need to build the CNC

Metric allen wrenches

16"x24" carpenter's square

12" rafter angle square

½" Plunge cutting router bit (to level the spoil board)

Circular saw (to cut the baseboard and spoil boards to size)

Precision screwdriver set (for lots of electrical screw terminals)

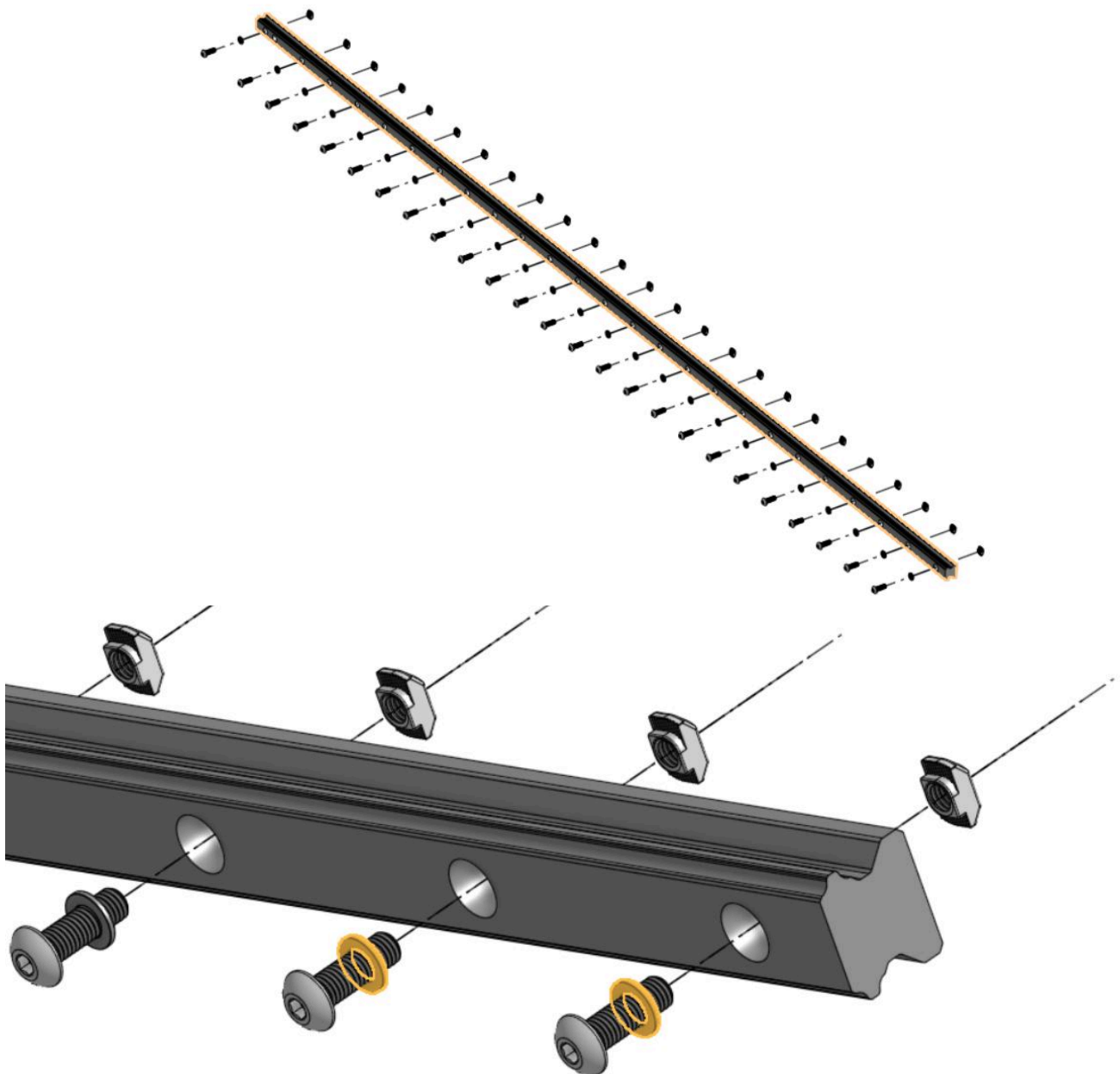
Crimper (for JST connections)

Soldering iron and solder



X-AXIS

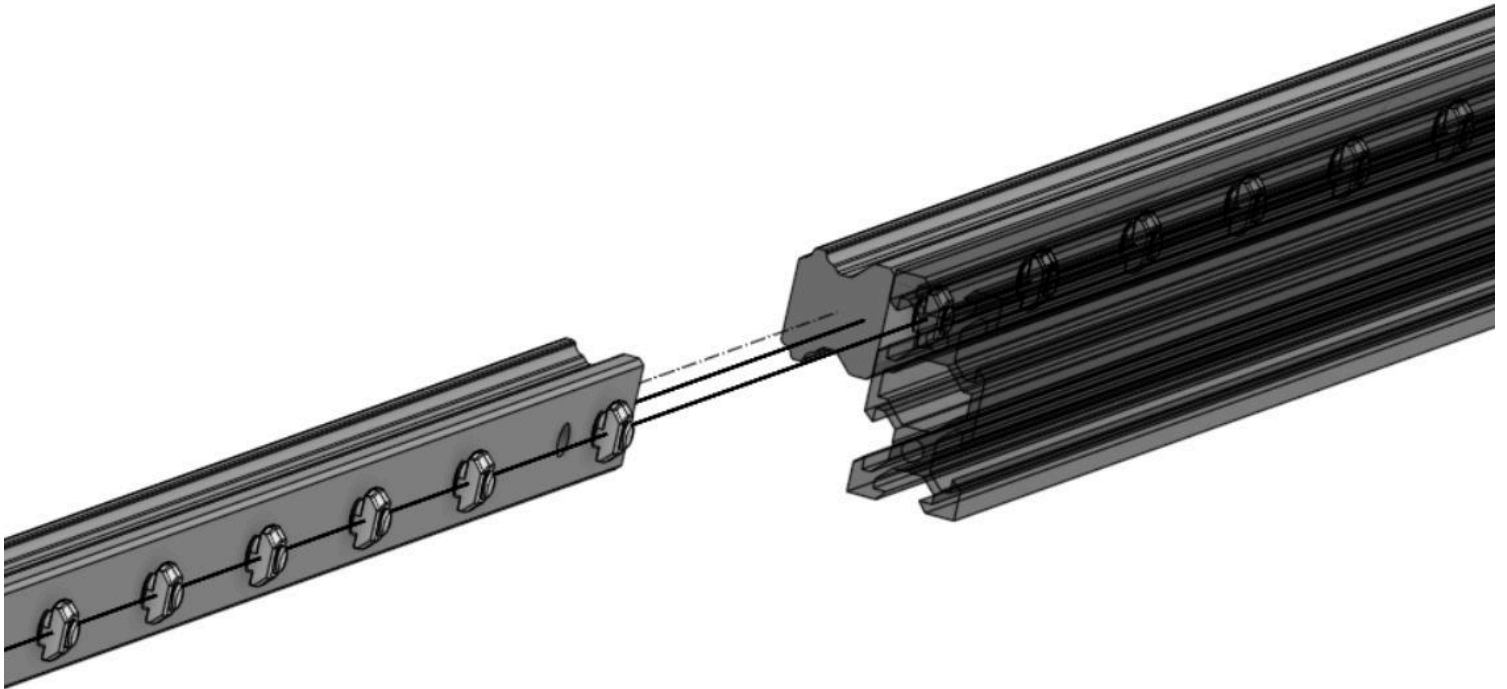
Installing the Rails



Add M5x16 Screws along the rail with split Lockwashers underneath.
Fasten with a M5 T nut, but do not tighten.

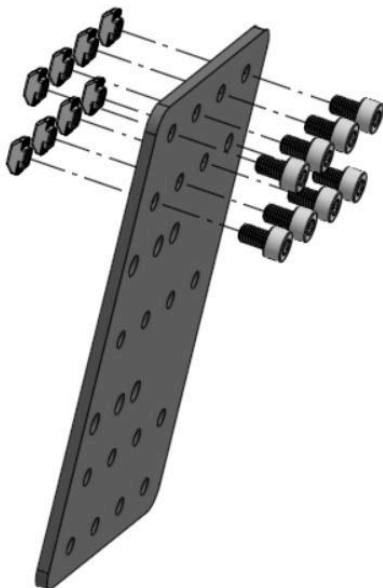


Installing the Rails



Slide the HGR20 rail onto a side of the 2040 bar.

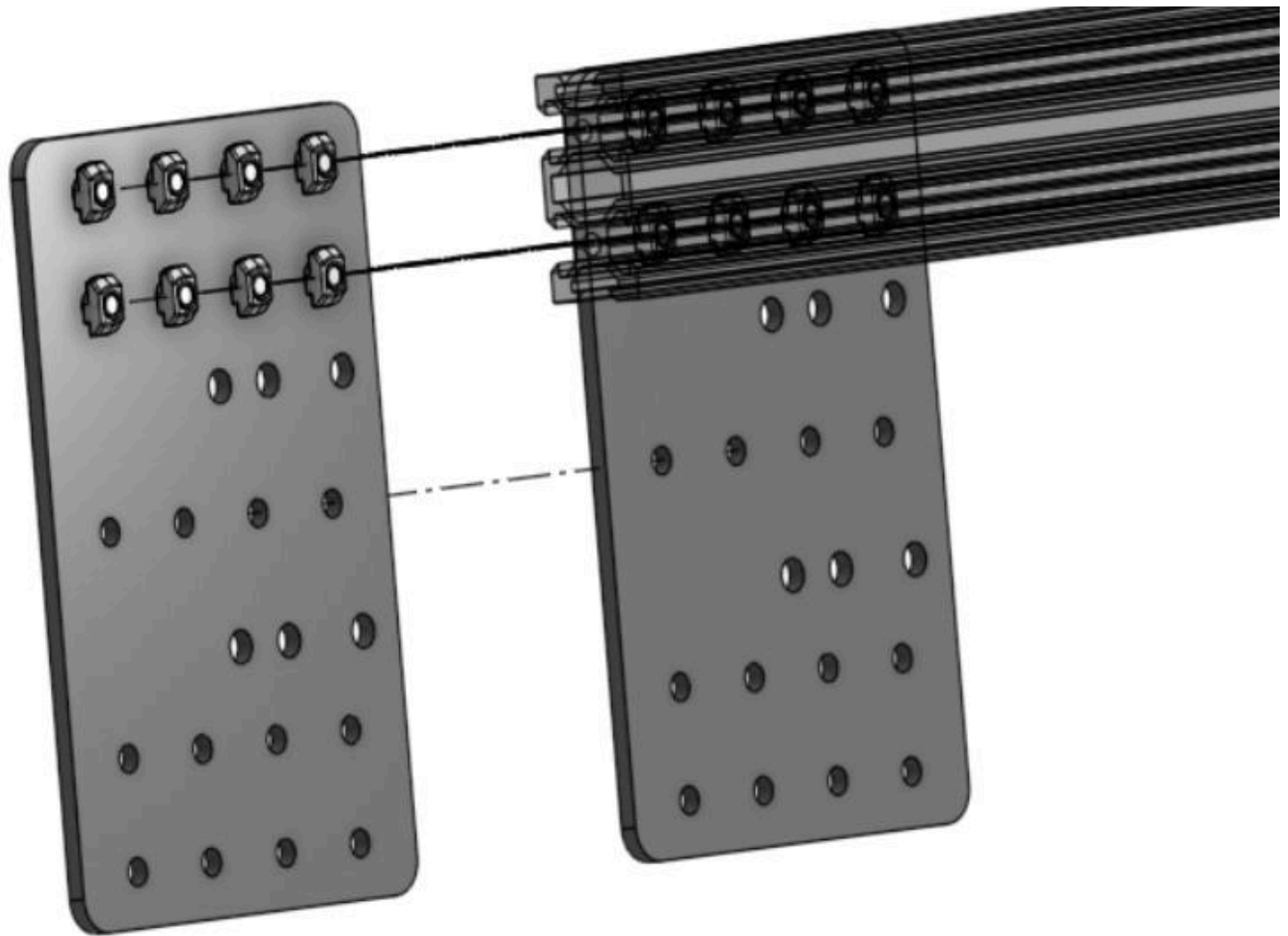
Omniplate Assembly



All omniplates for the X-Axis use the same assembly (M5x8 Screws and M5 T nuts as shown). Do not tighten the M5 T nuts.



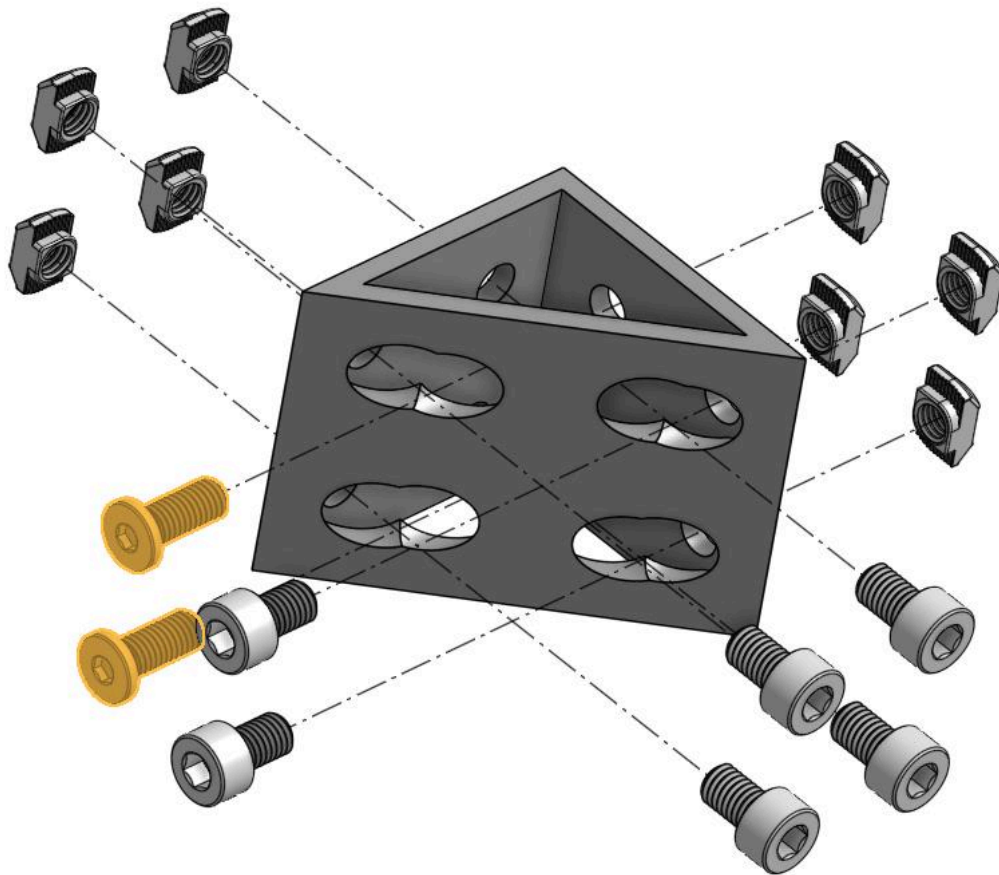
Omniplate Assembly



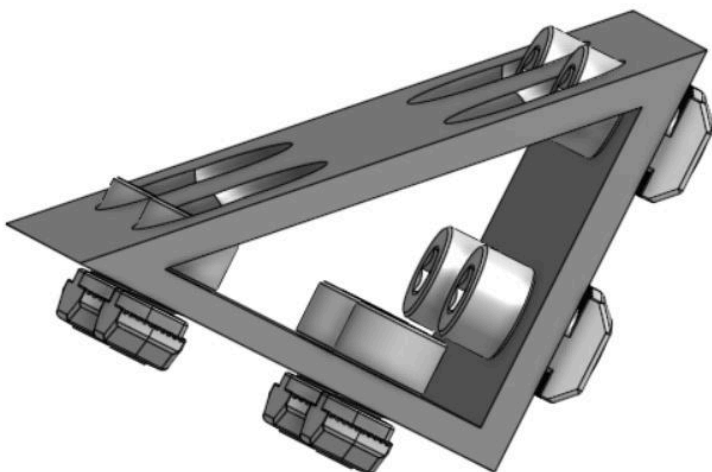
Slide the assembled omniplates onto the 2040 Aluminum profile(1500mm), and tighten the M5 Screws.



Corner Bracket



Prepare the corner braces as shown. Some holes require ultra low profile screw heads, follow the example. Use M5x8 Screws where not using low profile screws. Add M5 T nuts to the back of the brace, but do not tighten.



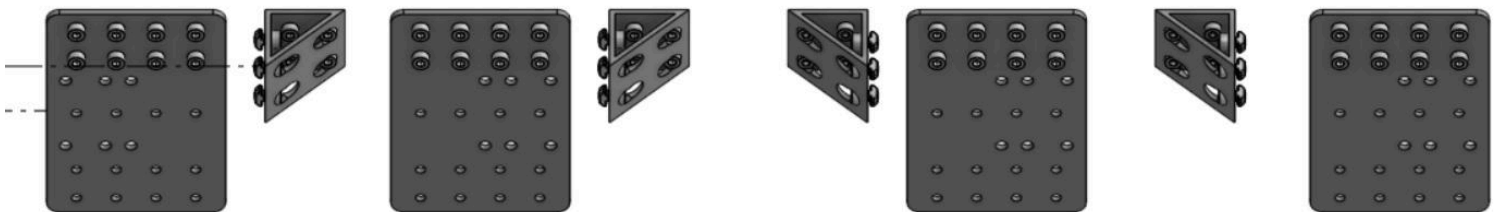
The corner braces are for spoil board support. This image does not contain ultra low profile heads, but most corner braces will need them in the inner corner.



Corner Bracket



Slide the Omniplates(x4) and corner braces(x4) as shown by the diagram (sequencing also shown below).

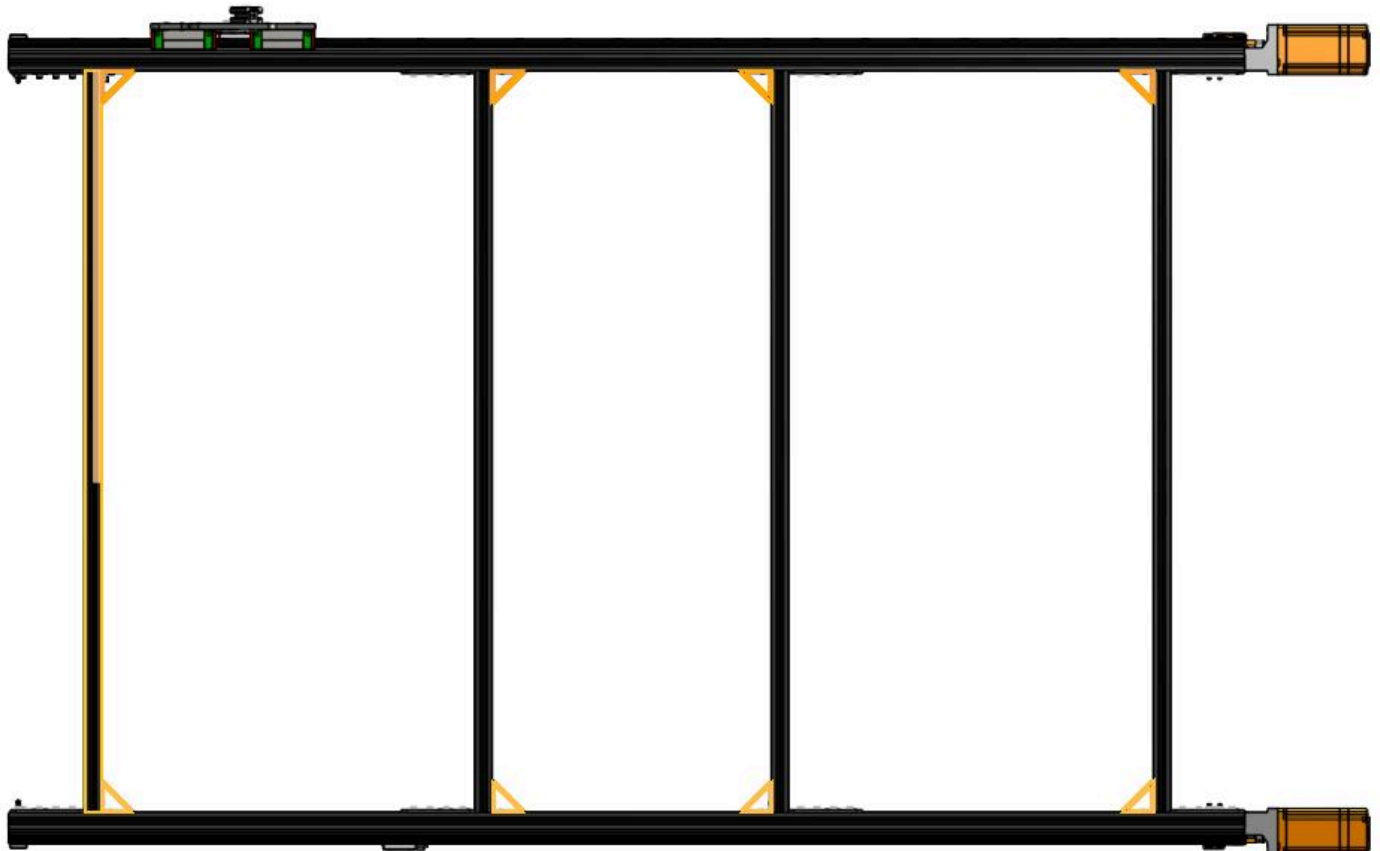


After sliding all plates and braces on, leave them. Do not screw them down until later steps.

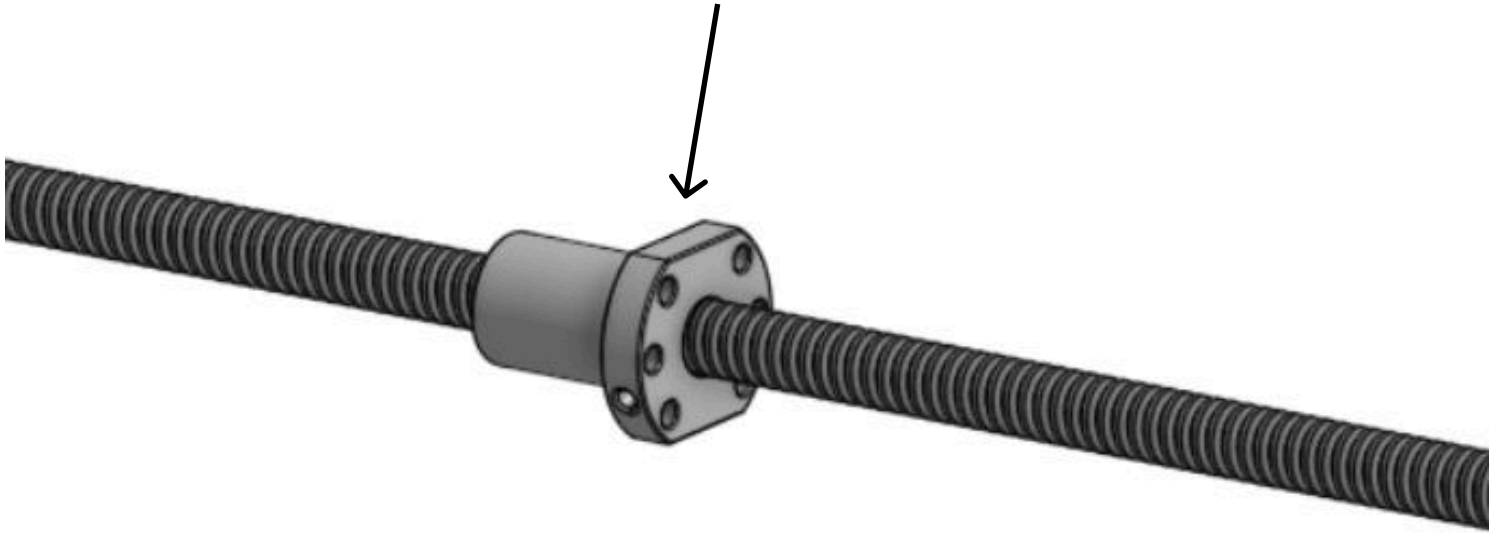


Corner Bracket

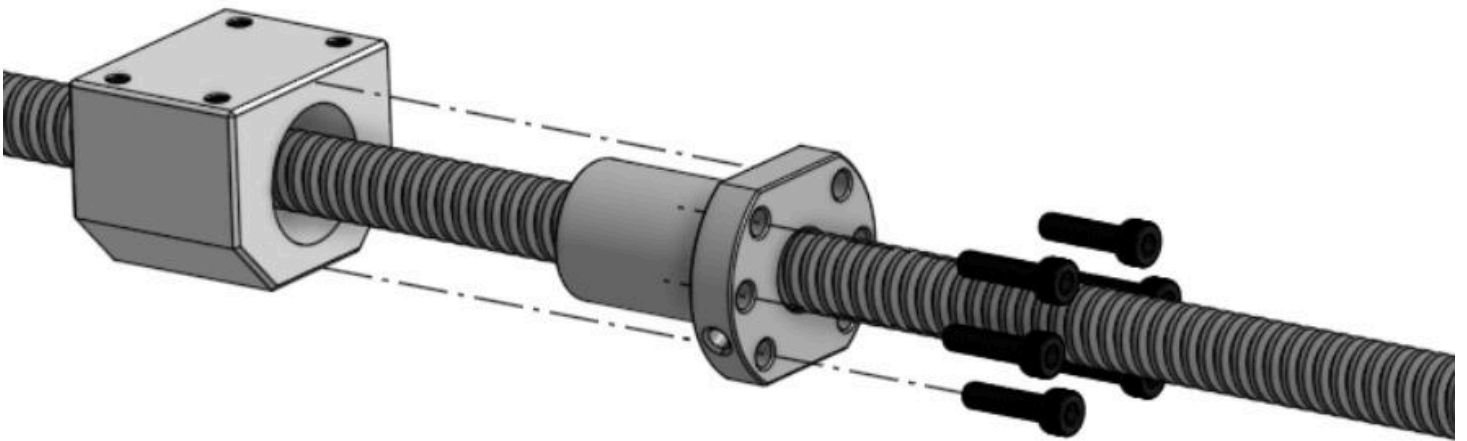
When making the second X-Axis rail, make sure the corner braces are aligned. In a future step there will be 2040 Aluminum profiles that connect the axes.



X-Axis Ball Screw (this is PRE ASSEMBLED SO DONT TAKE APART)



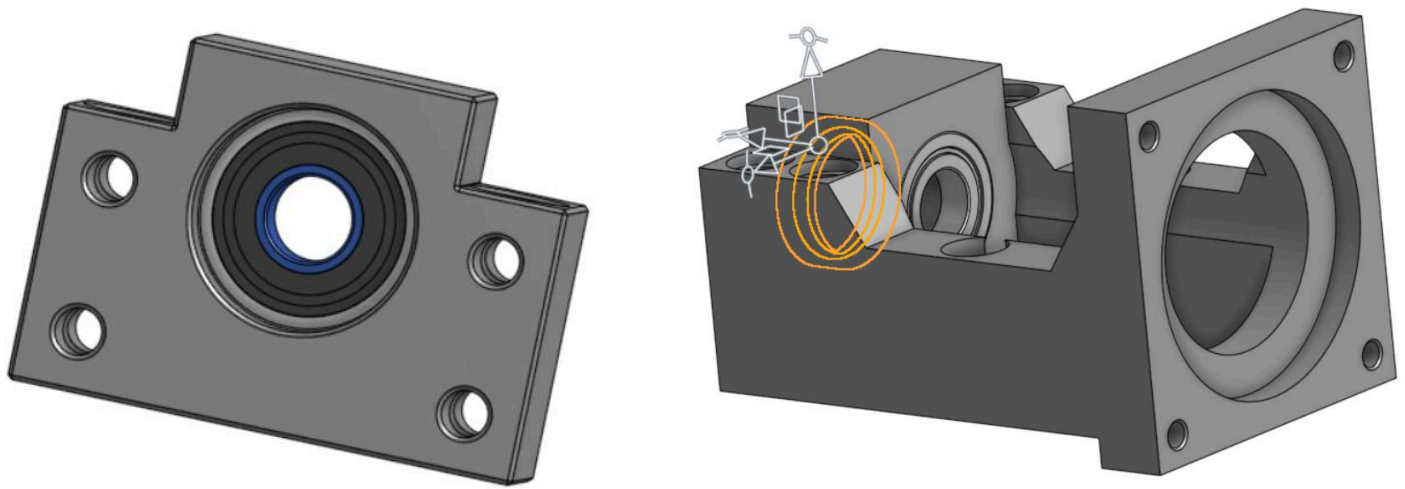
Ensure that the Ballscrew is in the center of the leadscrew, it will make later steps easier.



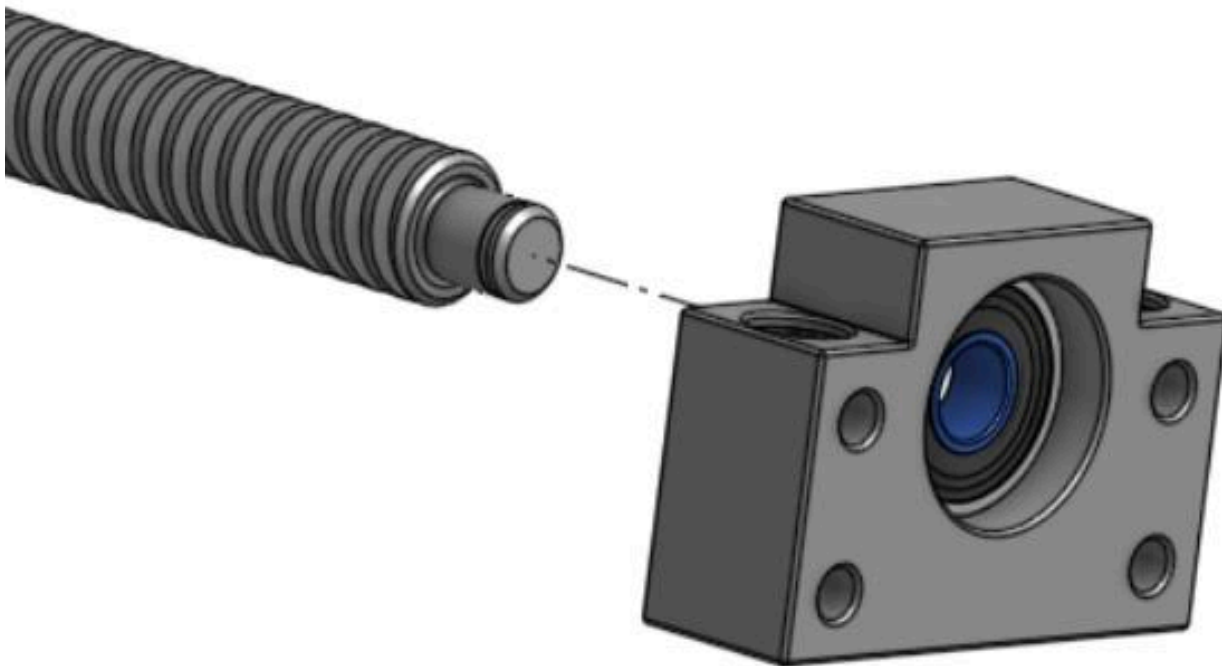
Attach the DSG16H Ball Nut with M5x20 screws as shown(You can also use 4 screws instead of 6). There are screw holes in the actual DSG16H Ball Nut Housing.



X-Axis Ball Screw



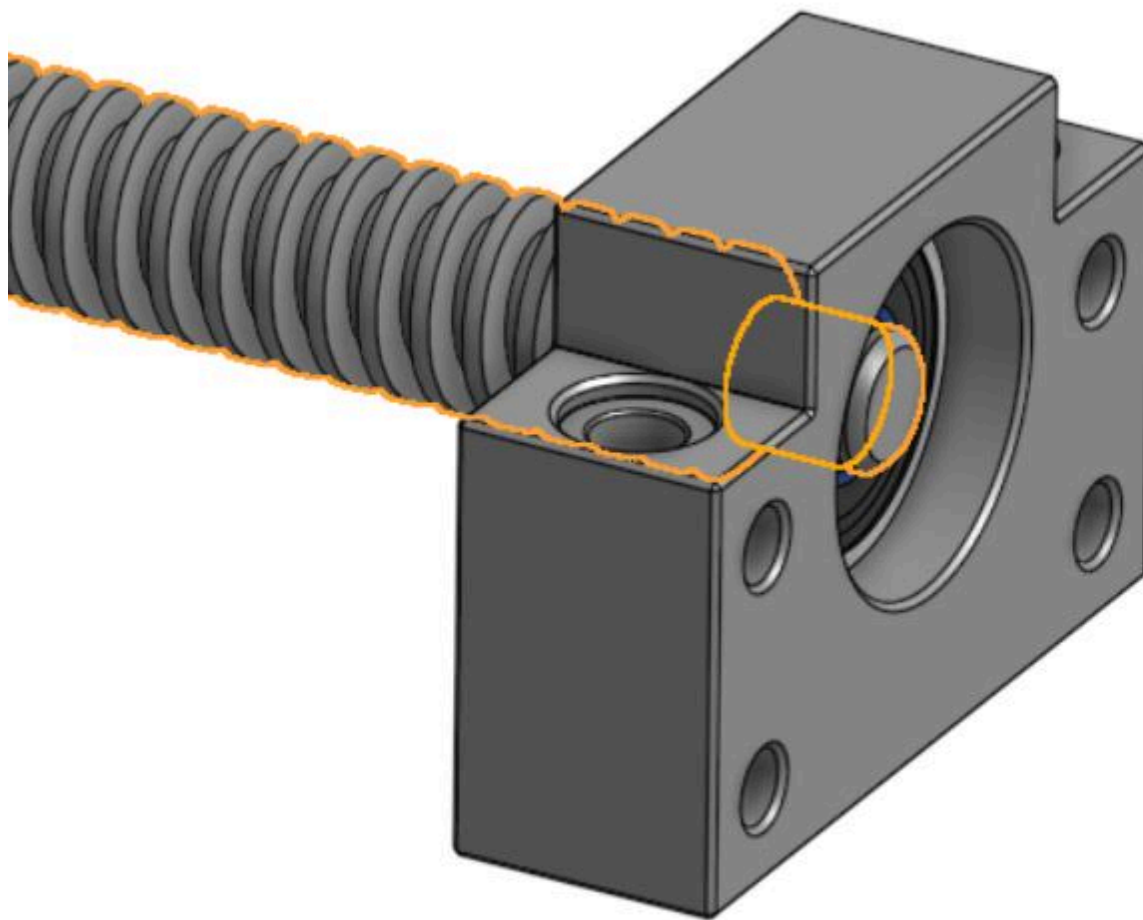
The BF12 and HM12 should already have bearings in place.



Slide the BF12 into the non-threaded end of the leadscrew.



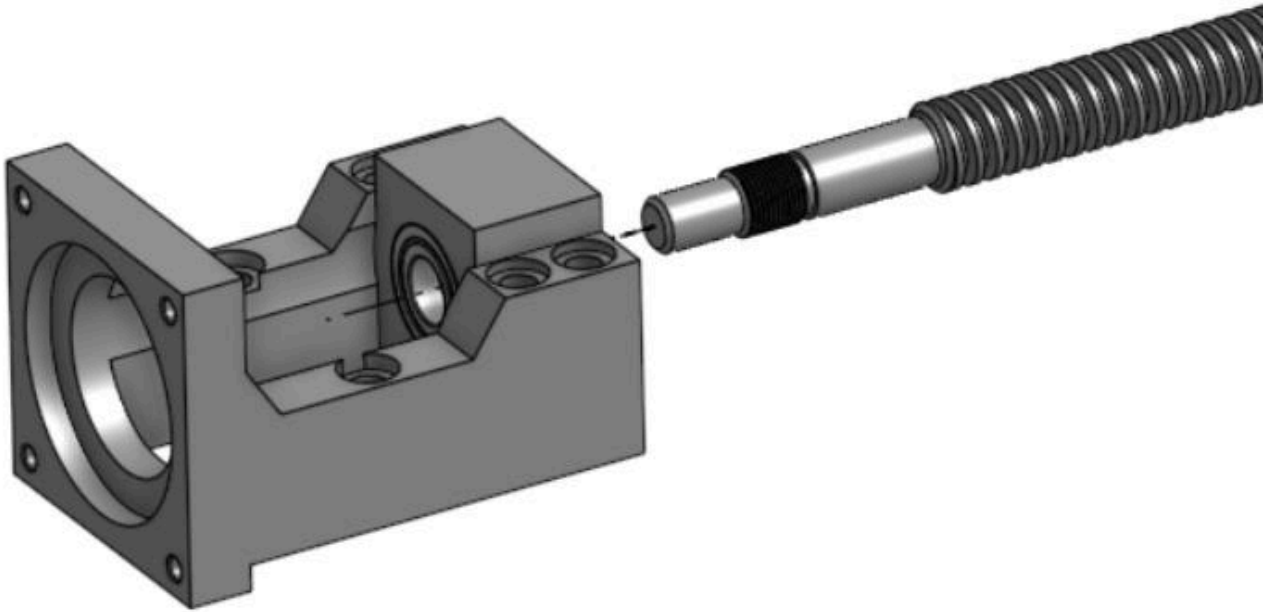
X-Axis Ball Screw



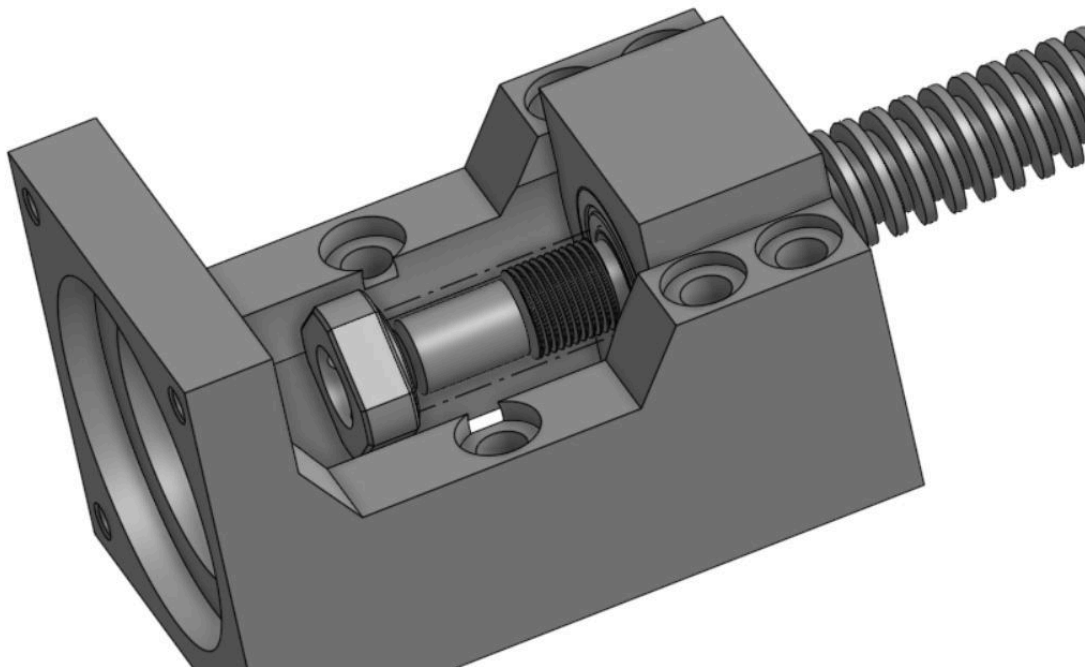
The assembly should look like this.



X-Axis Ball Screw



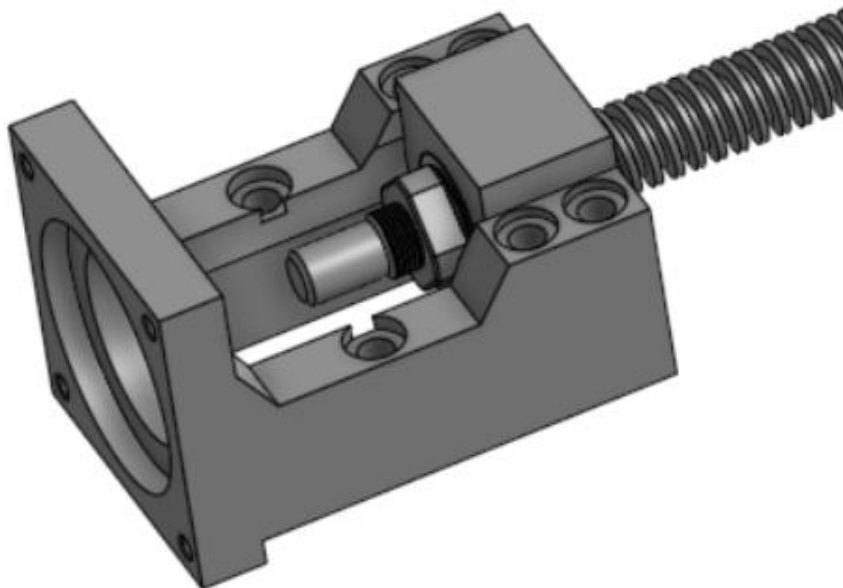
Slide the HM12 onto the other end of the leadscrew.



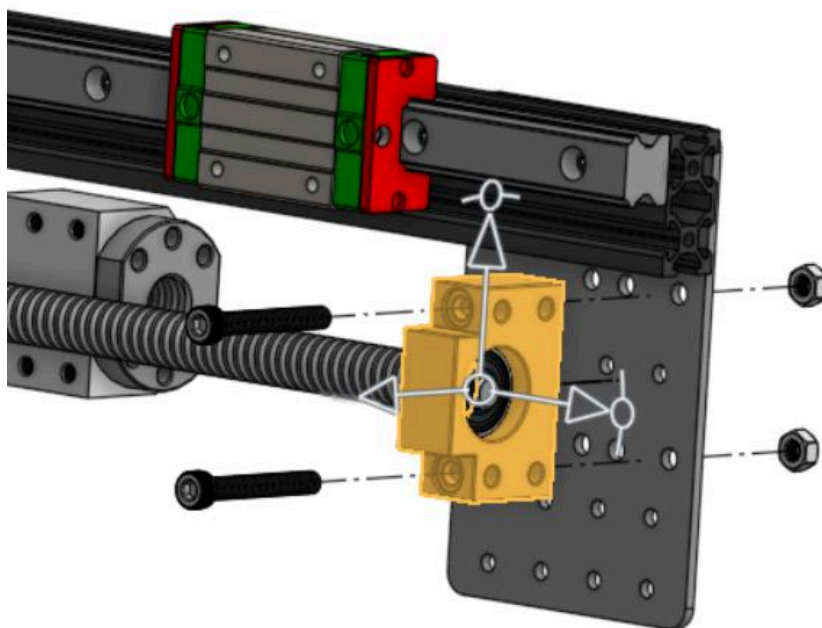
Screw the appropriate nut (comes with the HM12) onto the leadscrew as shown. Make sure it is tight! There's normally a setscrew on one or more corners of this nut. Tighten it to prevent the nut from loosening.



X-Axis Ball Screw



The finished assembly should look like this.

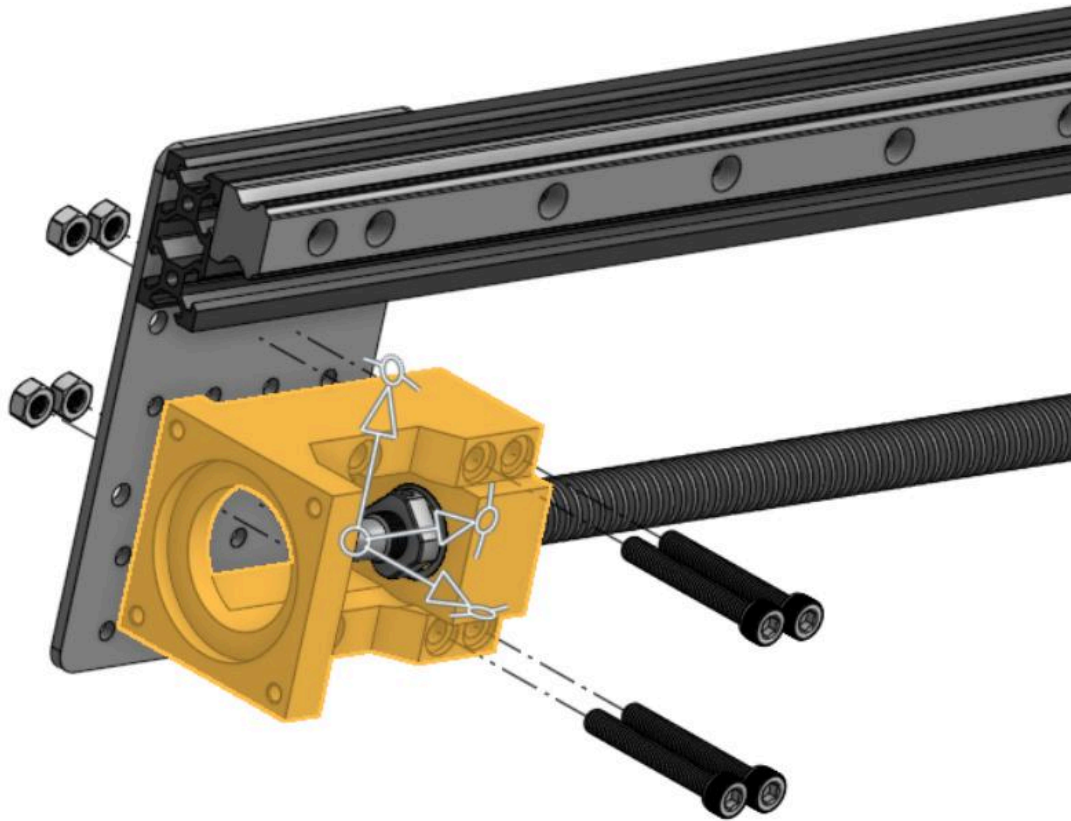


Make sure the BF12 is slid over the bearing and screw it onto the omniplate with two M6 nuts and two M6x40 screws



X-Axis Ball Screw

On the other side...



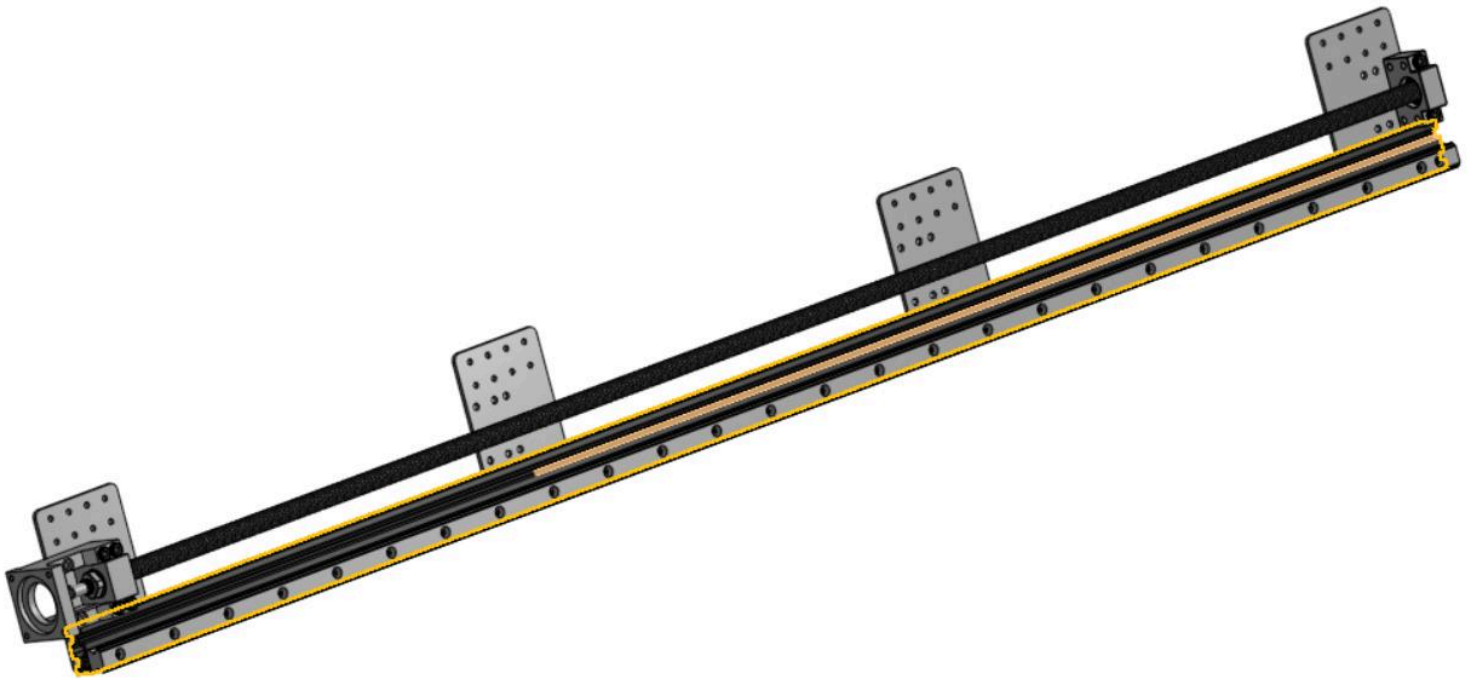
Screw the HM12 onto the omniplate using four M6x45 screws and M6 Nuts.



X-Axis Ball Screw

One of the X-Axis is now finished!

The second axis is assembled the same, but the HGR20 rail needs to be on the other side of the ball screw.



Assemble per the image (slider carriages will be on)

Now you are going to put the x-axis together.



Putting it Together

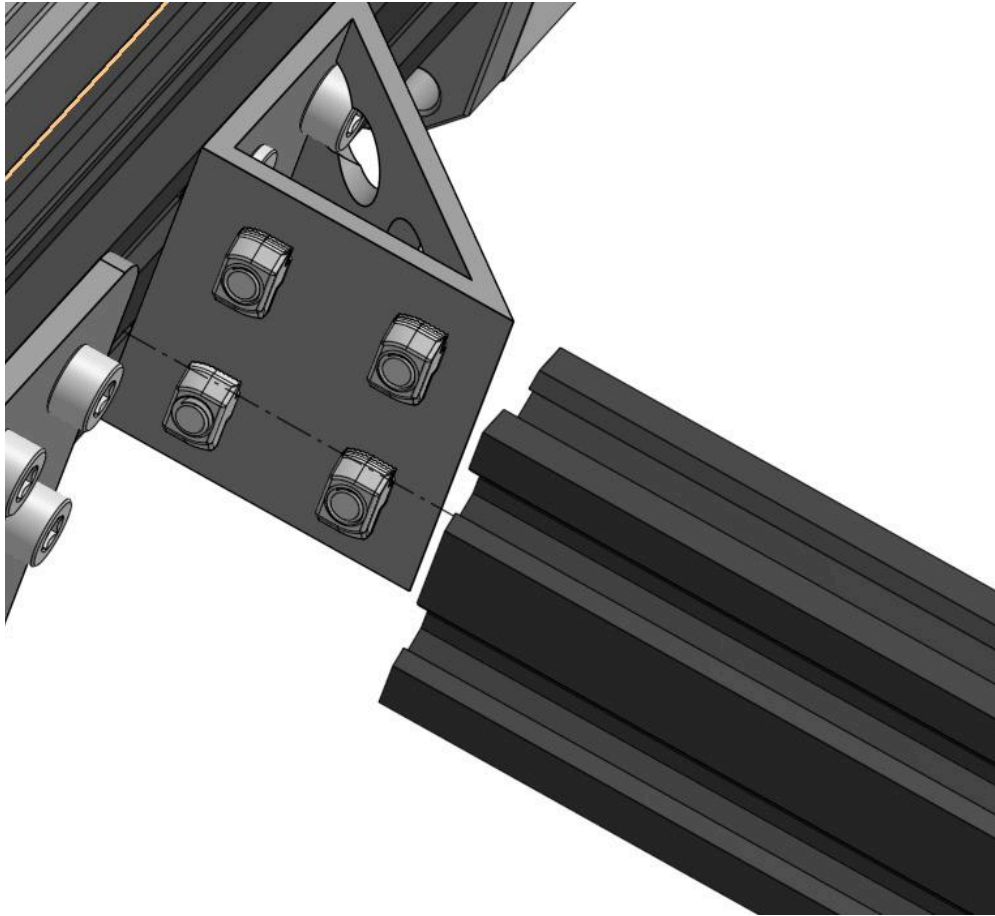


Slide on the 900mm long 2040 bars. They attach onto the Corner Braces.

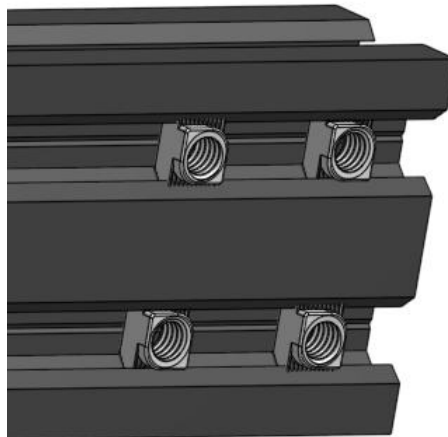
Tip: We bought 1000mm bars and cut them to 900mm. If you do the same, keep the ~100mm pieces, they get used later.



Putting it Together

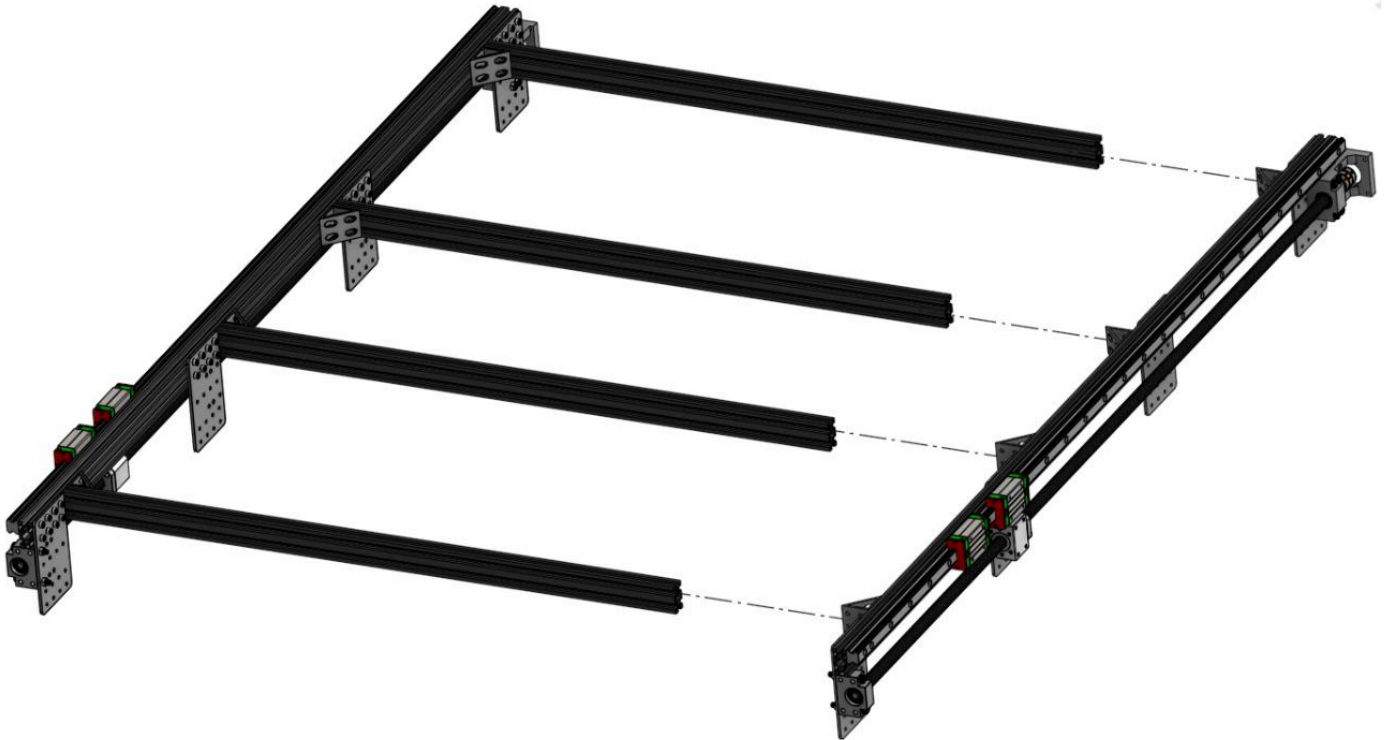


One way to make the assembly easier is to loosen the screws that work on the corner braces and main X-Axis assembly. Another tip is to put the hammer head nuts onto the 900mm 2040 bars. and align them with the screw holes prior to screwing it in.



Putting it Together

You are then going to put on the other part of the x-axis:

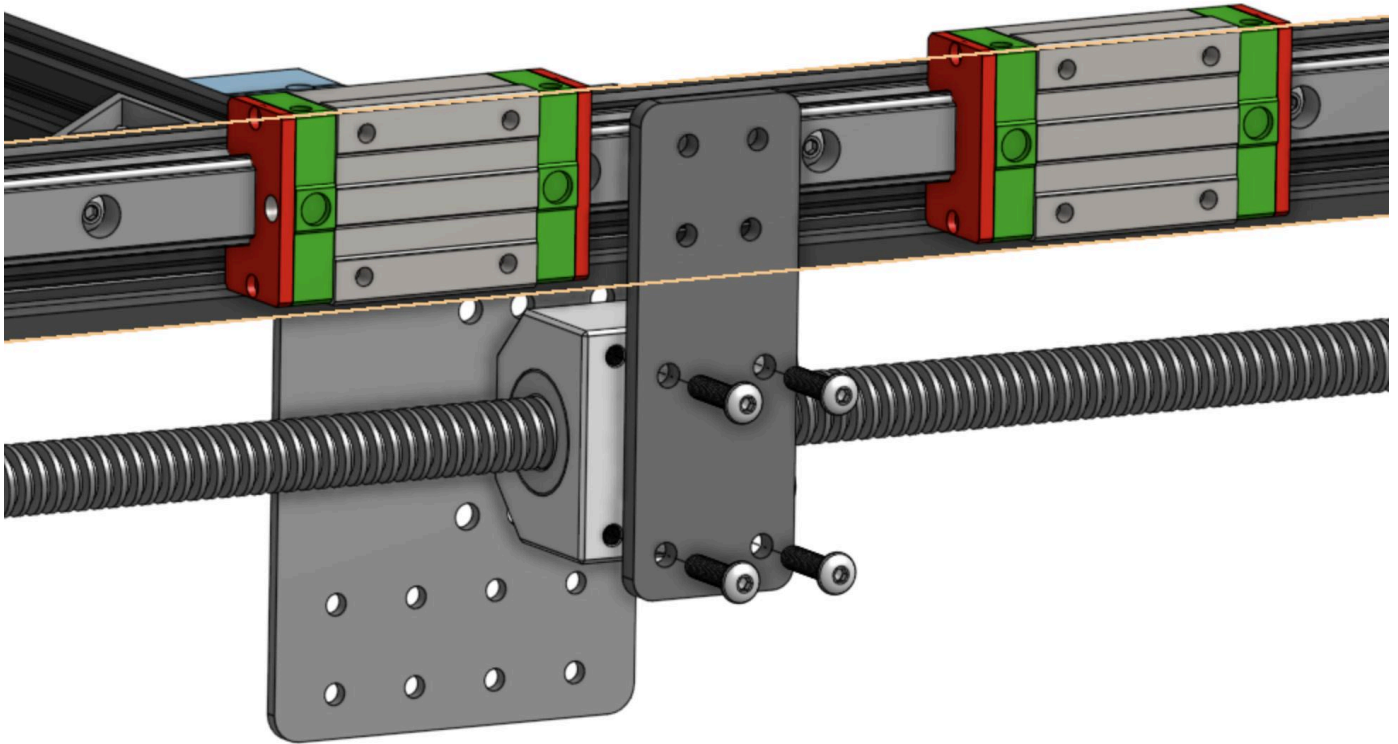


It is recommended to place the x-axis with the attached bars onto a table and use the same process as before to attach each of the corner braces to those 900mm 2040 bars.



Putting it Together

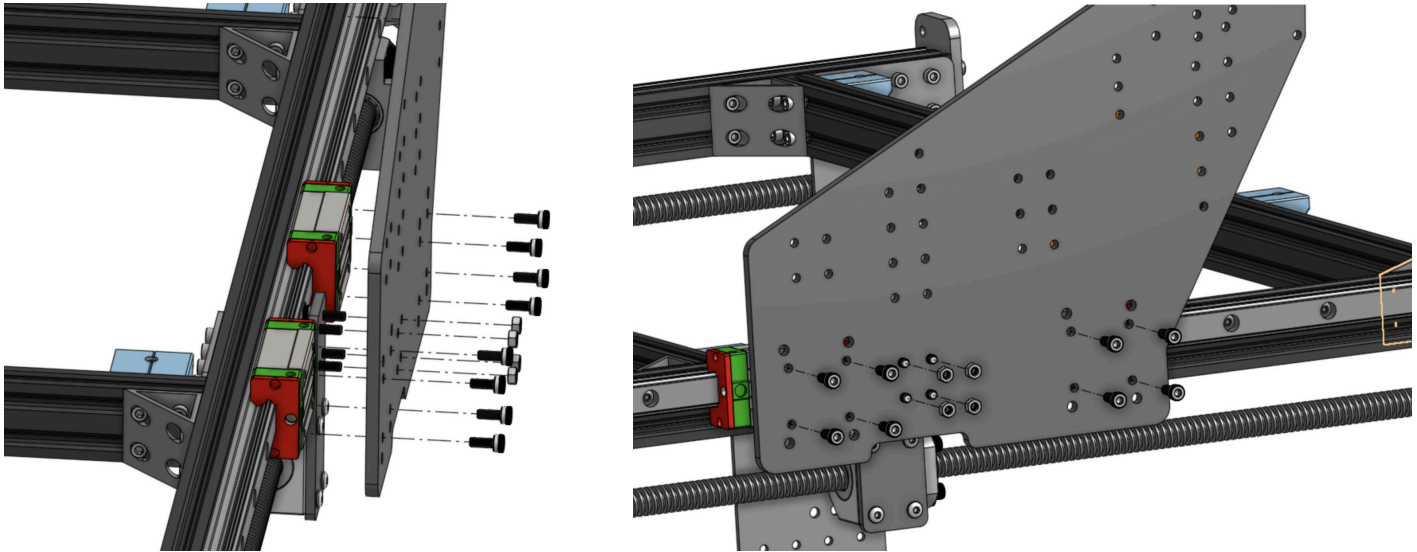
Next, grab the tie plates



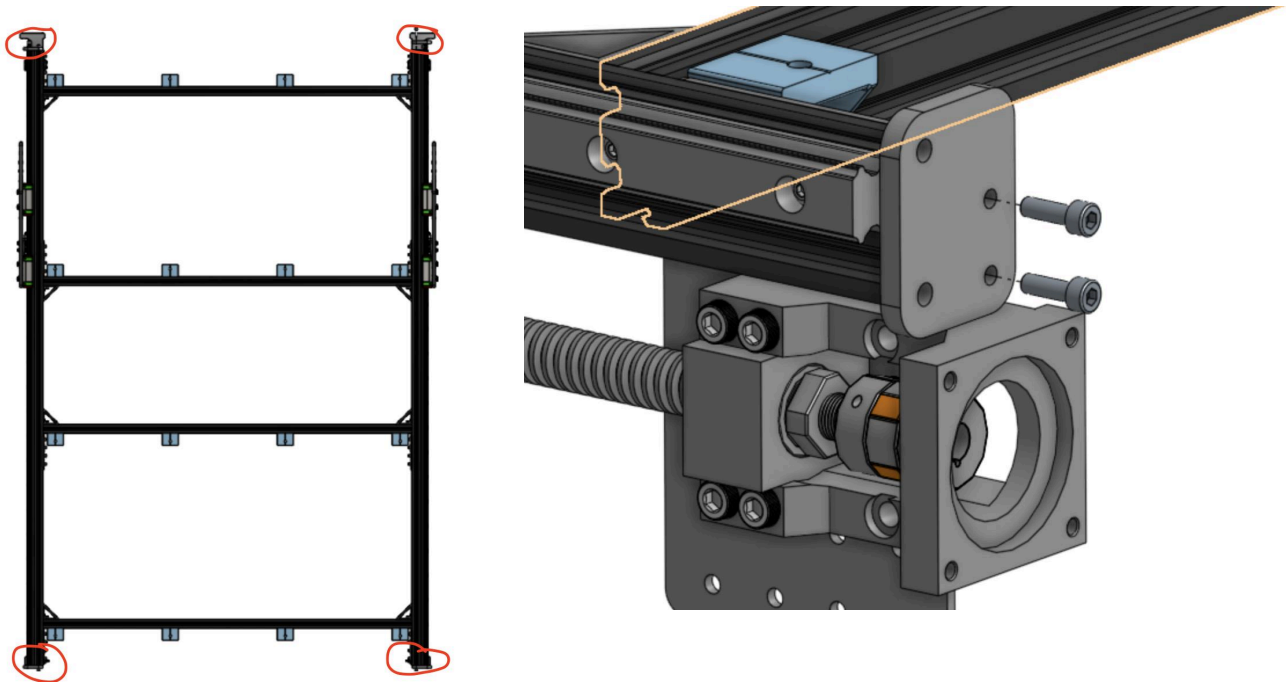
Screw the tie plates into the threaded holes of the DSG16H Ball Nut Housing using 4 M5x20 screws. **IMPORTANT: MAKE SURE THE TIE PLATE IS IN BETWEEN THE LINEAR BEARINGS.**



Putting it Together



Fasten the Gantry Plates to the Two Linear Bearings and the tie plate using eight M5x12 Screws (Linear Bearings) and four M5x16 Screws with four M5 nuts (Tie plates).



Finally, screw in the hard stop plates using two M5x0.8 screws. Repeat this for all corners of the CNC as shown.



End Result

After installing the 3D printed Base Board/Spoil Board hold-downs, the frame should look like this:



Y-AXIS

Introduction

The gantry plate design for FRCNC supports any one of three configurations:

- omniplate mount
- 6"x1" (x0.125" wall)<- confirm thickness with Evan)
- 6"x2" (x0.25" wall)

The omniplate configuration is by far the simplest and least expensive to build, but when we used this configuration we could easily cut wood or polycarbonate, but the Z axis could flex under load and we struggled to cut aluminum). The 6x2 tube is much stiffer and handles aluminum easily. The main challenge of going straight to the 2x6 configuration is the level of precision needed for the holes in the tube (not easy without a CNC). If you have a good (precise) way to cut a long, heavy 6x2 then we suggest going straight to that option. If not you can do what we did and build the omniplate version first, then cut the 6x2 (with a really slow feed rate) and upfit your FRCNC.

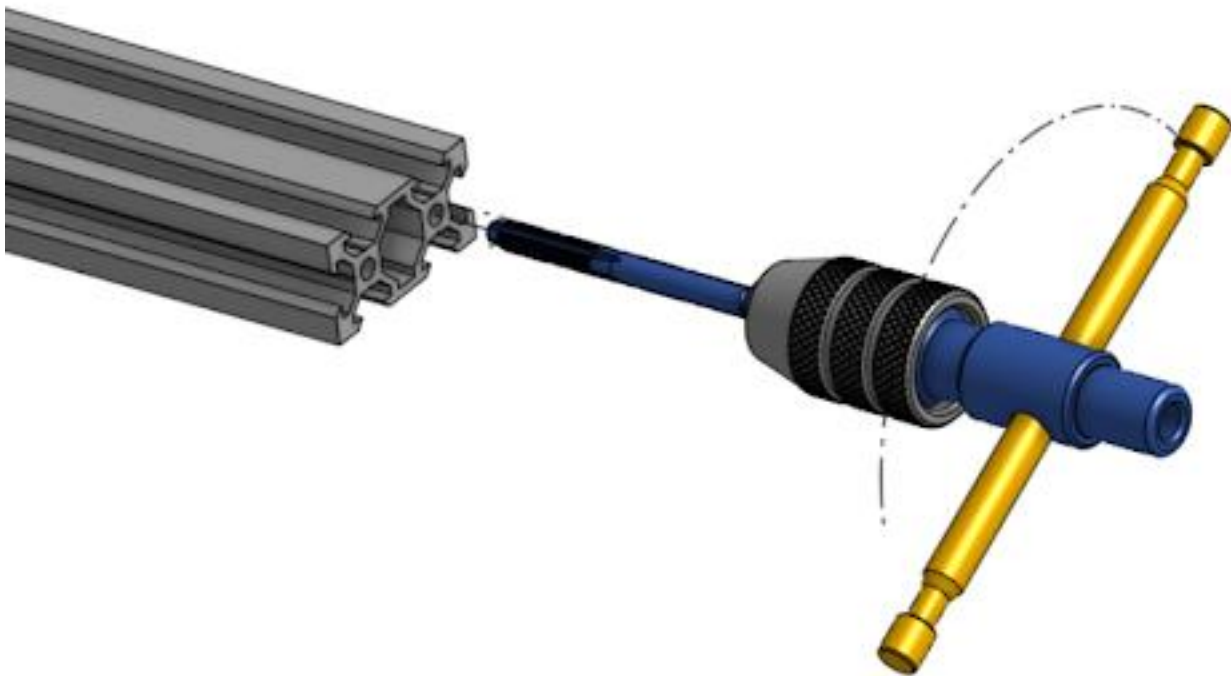


Y-AXIS

OMNIPLATE VERSION

This is the version we ran with for about 1 year, we call it our “V1” FRCNC because we’ve moved to a 2x6” Tube for V2

Preparing the 2040 Bars



Use an m5 tap to prepare the holes on the sides of your 1 meter long 2040 bars. Make sure this tap goes at least 30 millimeters, as that is how long the screws going through these will be.



Add the rails



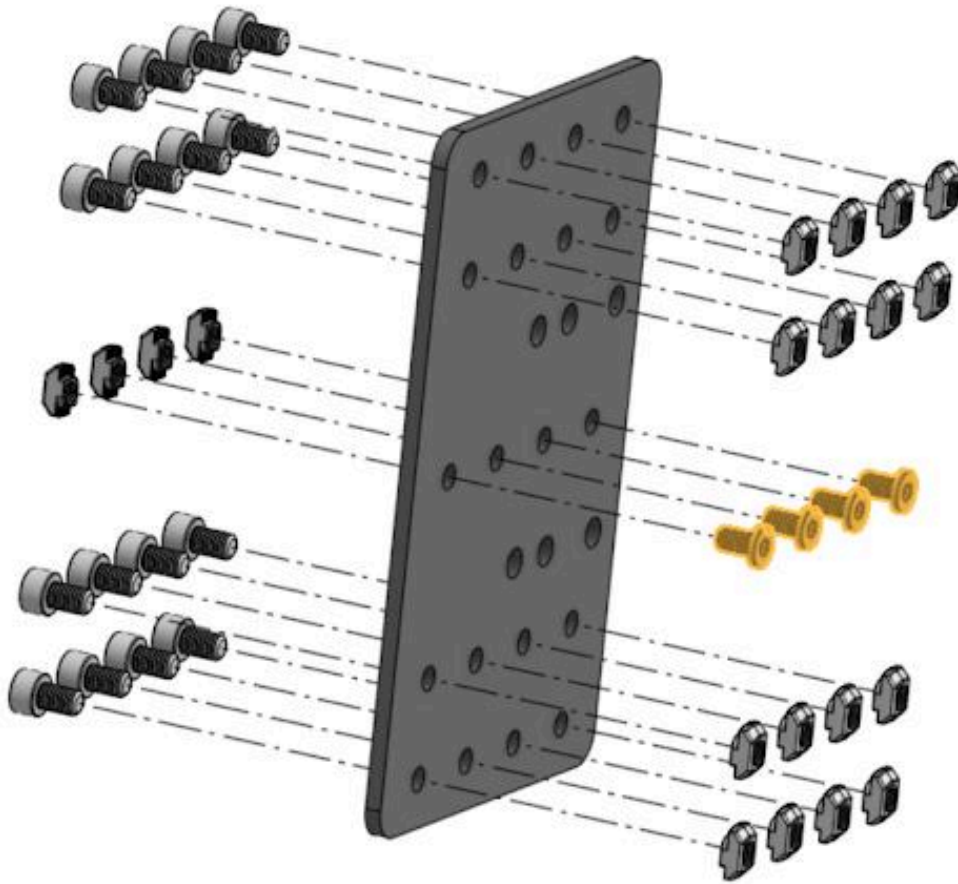
You will proceed with the same steps for the rails (these 2040 bars and rails are 1 meter long).



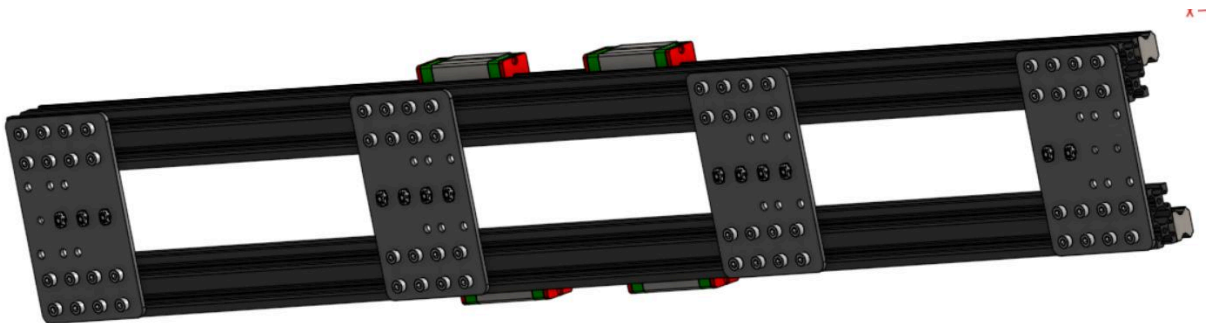
It should look like this.



Add the rails



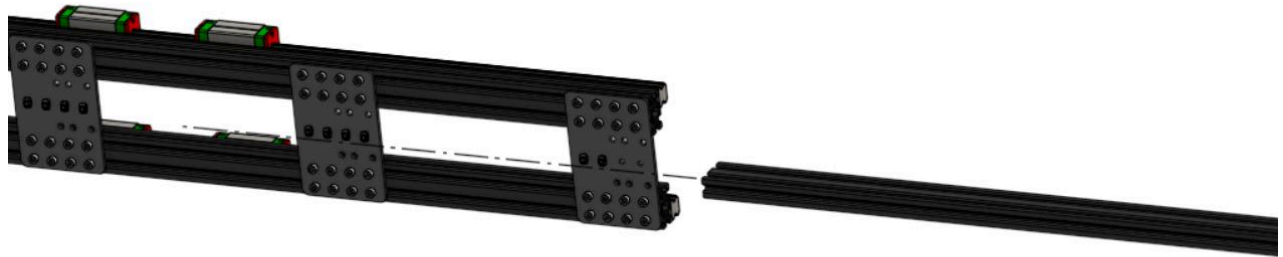
Here is how the omniplates are assembled. Same Screws and Nuts, but you need more as you can see. Keep in mind the holes in the middle, as the screws will be facing a different direction, and are ultra low socket head M5x8 screws.



Slide on these omniplates. Look closely at the two outside omniplates because a few of the middle screws need to be omitted for HM12 and BF12 to fit.



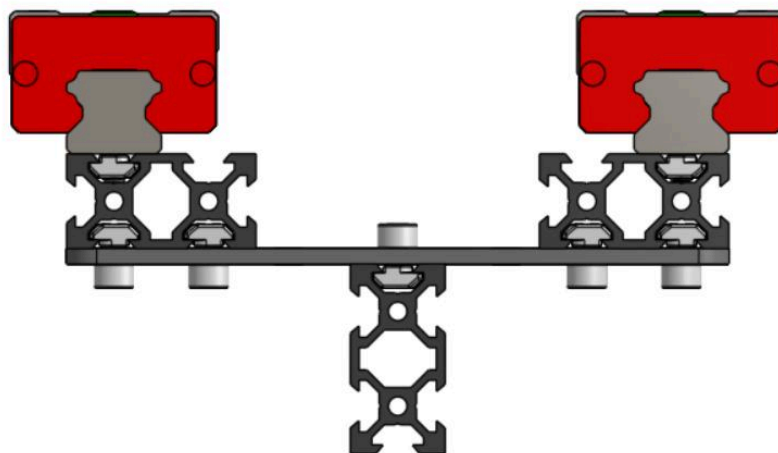
Add the rails



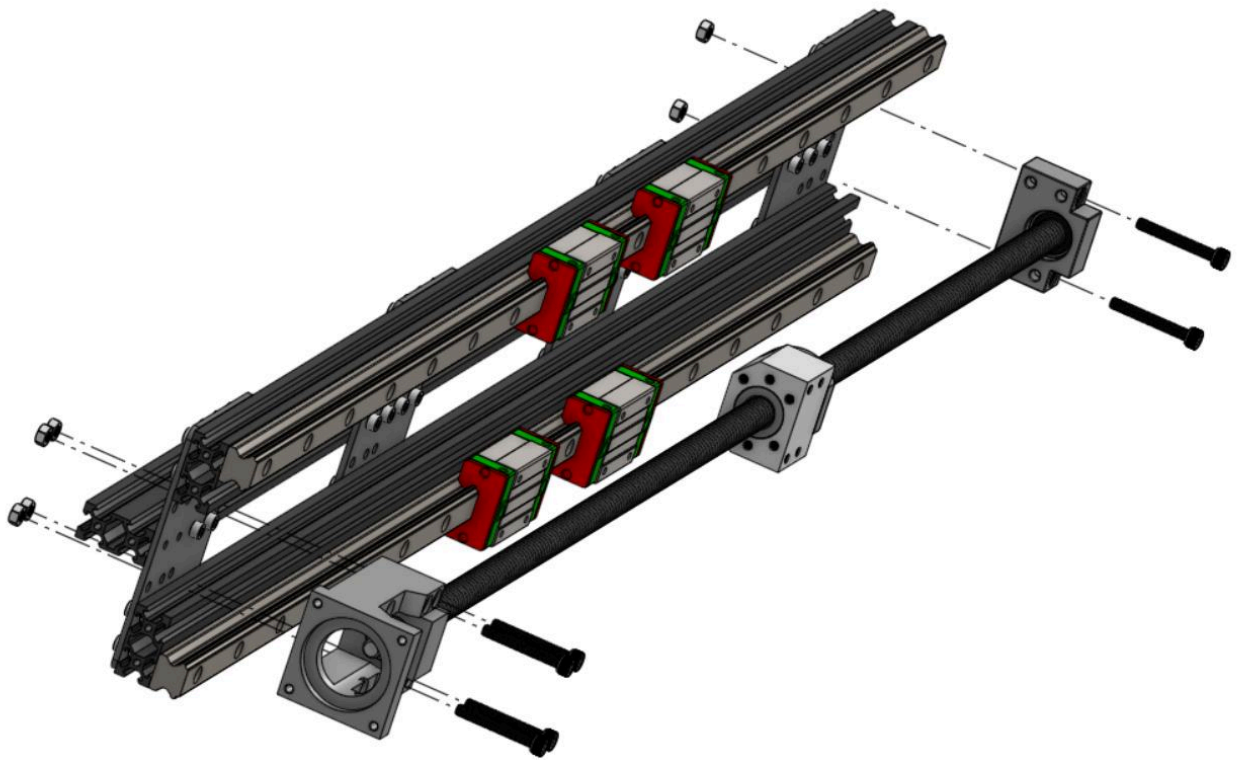
Slide in a 2040 bar onto the back of the assembly using the screws and nuts you put in the middle.



Top View

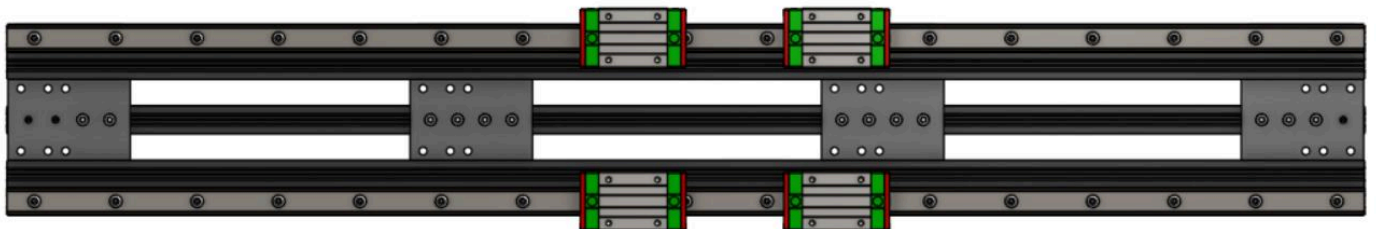


Add the rails



Now follow the exact steps you used to assemble the ball screw on the x-axis, and screw it onto the omniplates of the x-axis.

Make sure that the DSG16H and SFU-1605 are kind of in the middle of the ball screw because it will make assembly easier later.



Front View

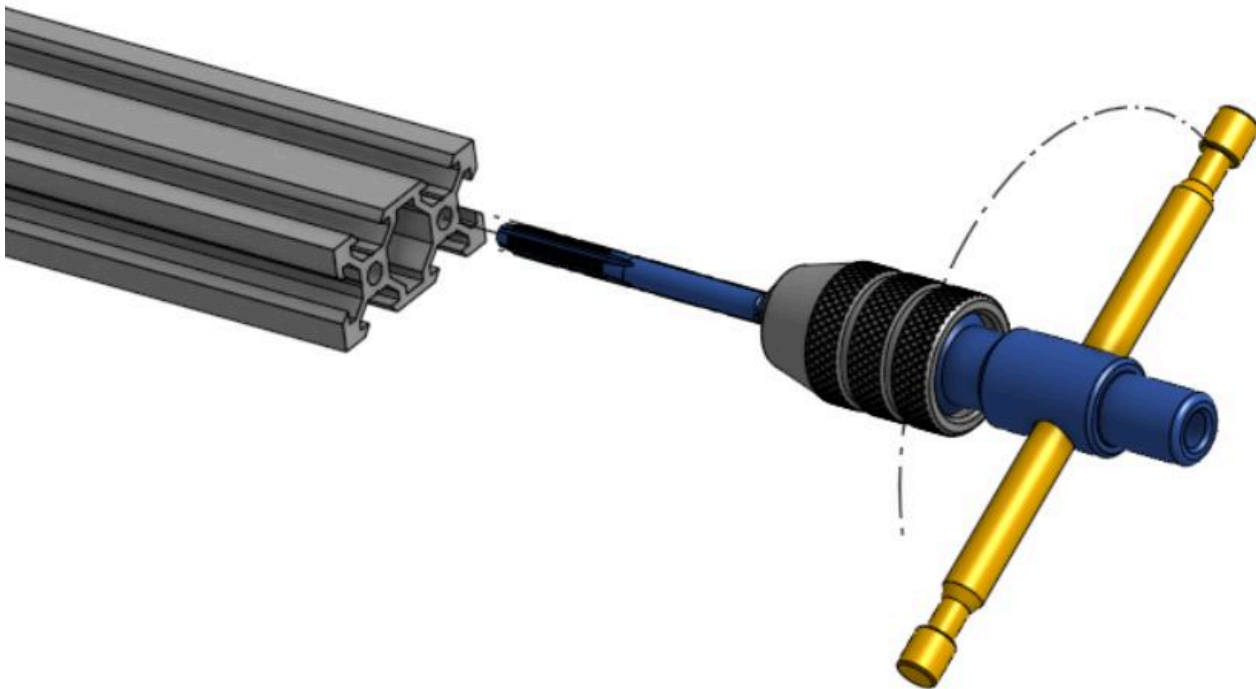


Y-AXIS

2X6" / 1X6" TUBE VERSION

This is the version we run now. We call it "V2" FRCNC.

Preparing the 2040 Bars



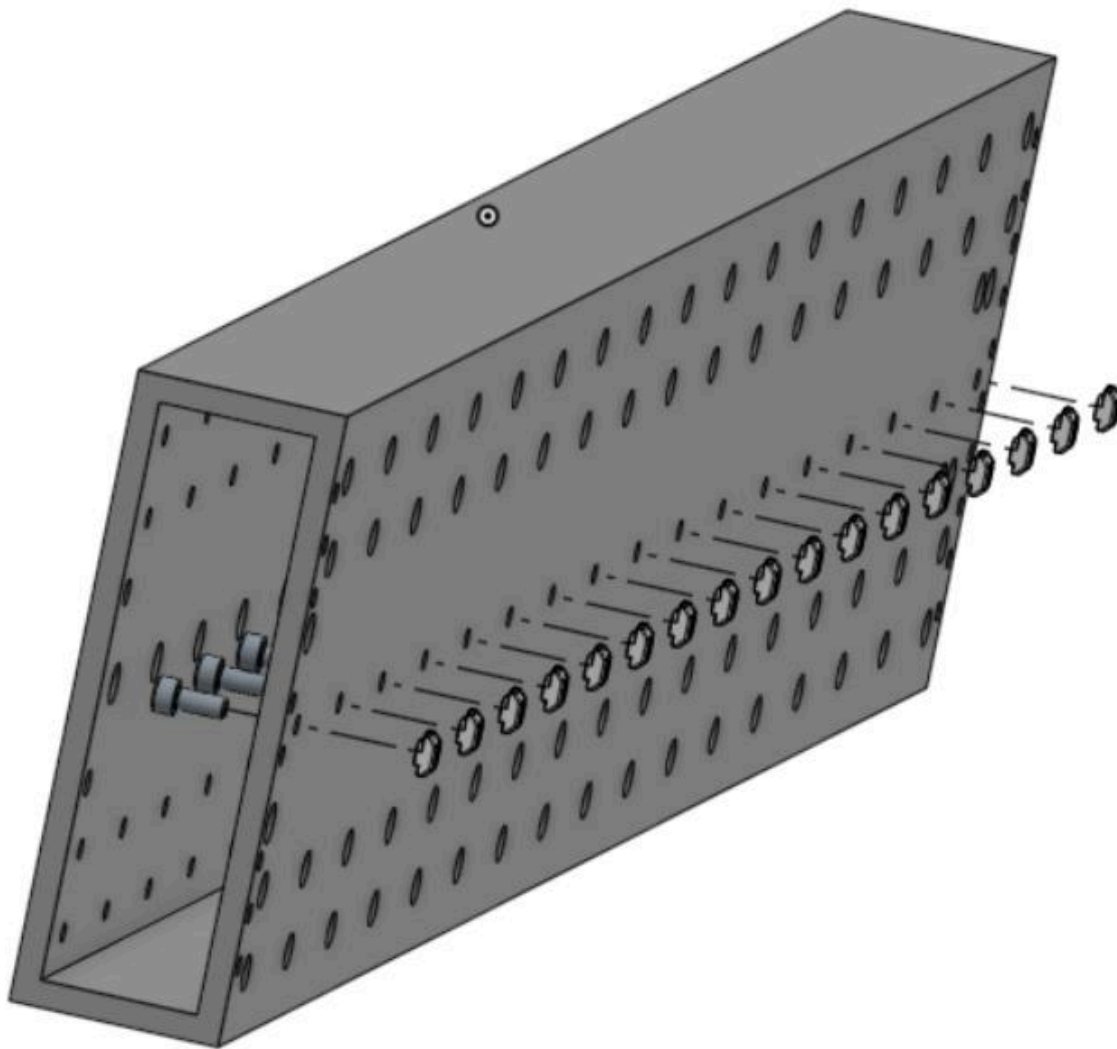
Use an m5 tap to prepare the holes on the sides of your 1 meter long 2040 bars. Make sure this tap goes at least 30 millimeters, as that is how long the screws going through these will be.



Add the rails

First, add an 1000mm long 2040 bar to the back.

Add 12mm m5 screws to the middle back row of holes.



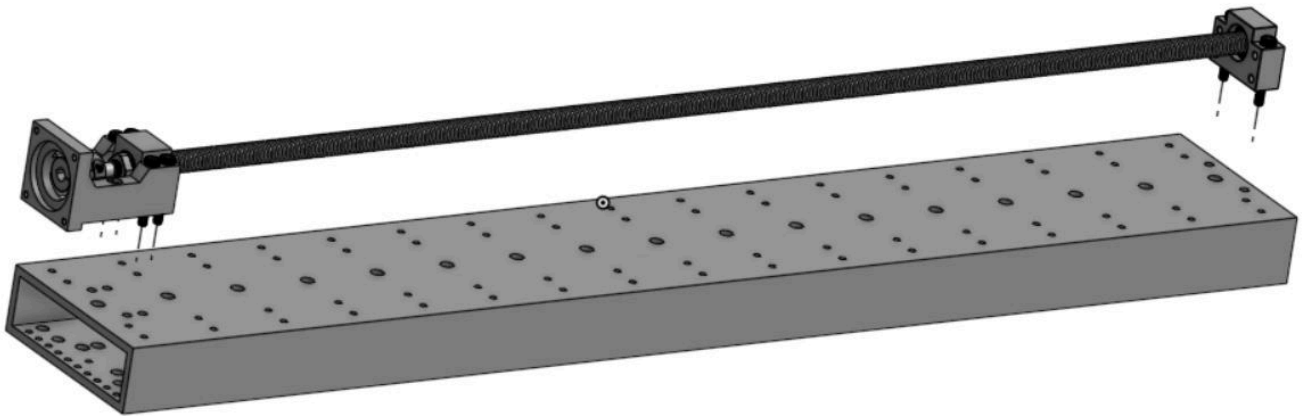
Add the rails

Then slide on the 2040 bar and tighten the screws from the front.

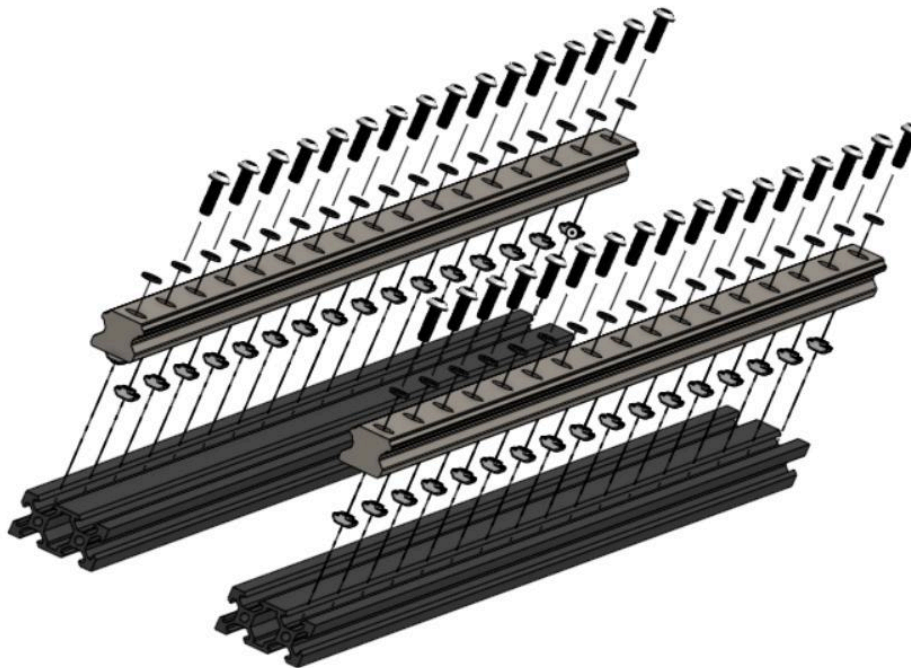


Add the rails

Add the Ball Screw System to the bar you are using.



Then prepare the Linear Guide Rails.



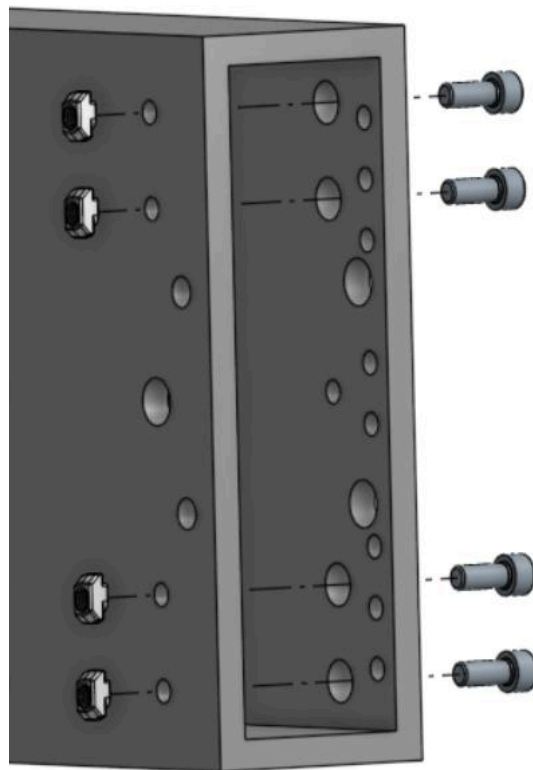
You will proceed with the same steps for the rails (these 2040 bars and rails are 1 meter long).



Add the rails



It should look like this.

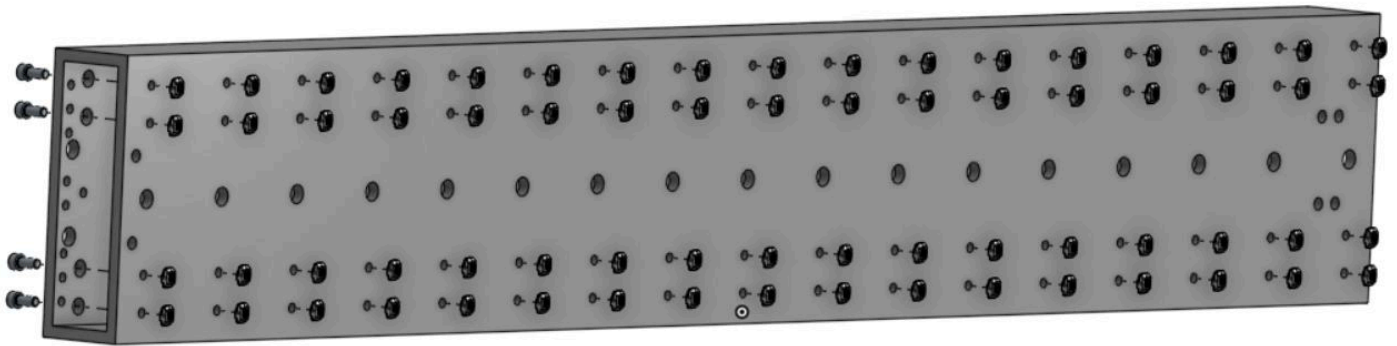


Add 5mm screws to the back of the tube you are using with T nuts on the other side.

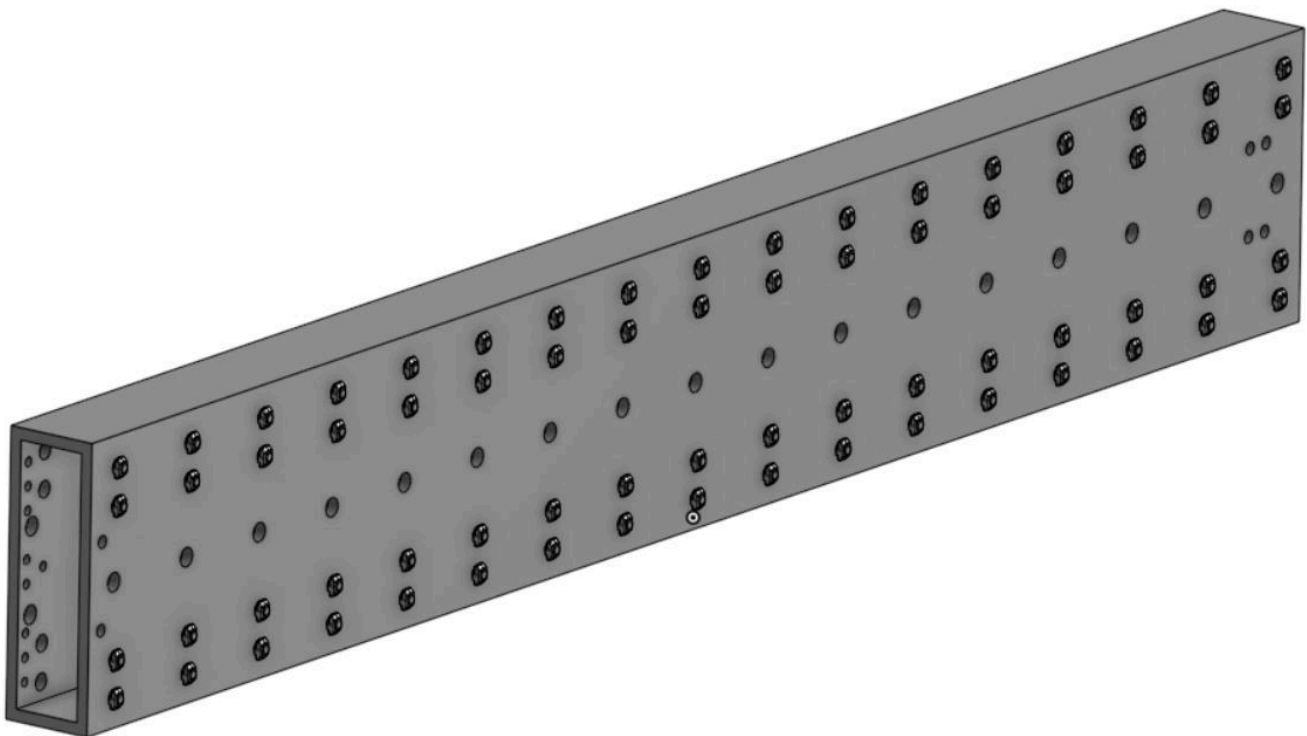


Add the rails

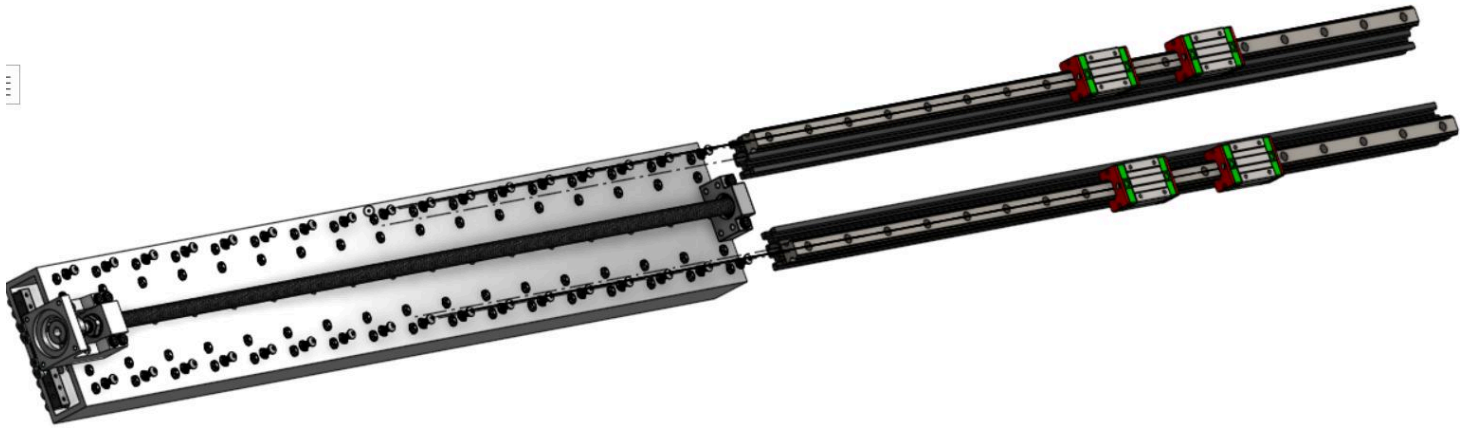
Add 5mm screws with T nuts for all the holes you see below.



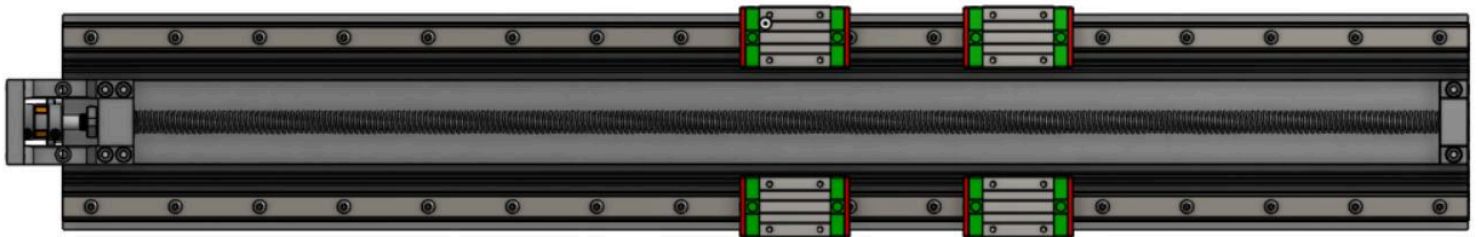
It should look like this.



Add the rails

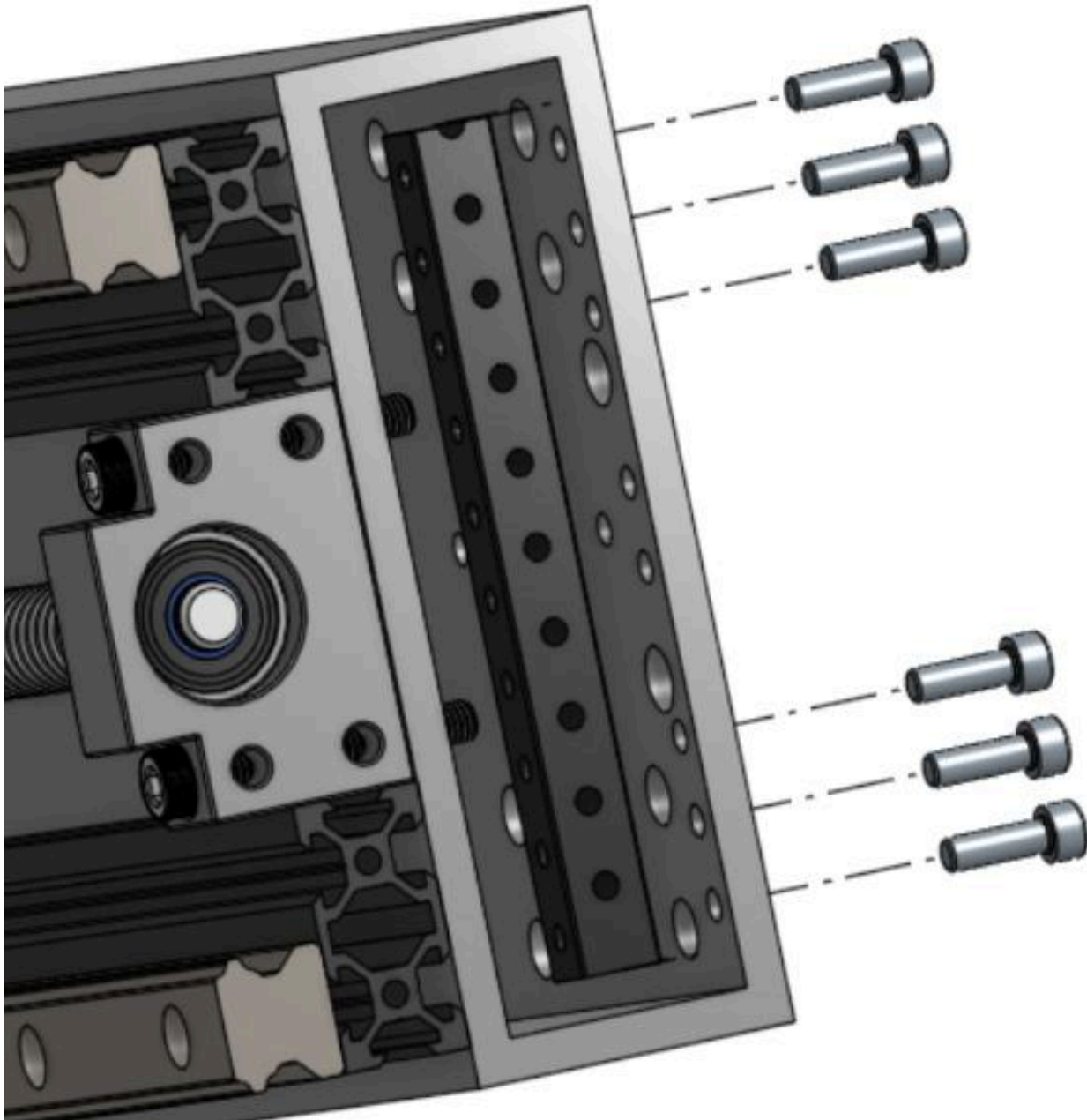


Slide on the linear guide rails, align the edges of the tube, and tighten the screws on the back of the support bar.



Add the rails

Add a .5" thick 10-32 WCP nutstrip to both sides:

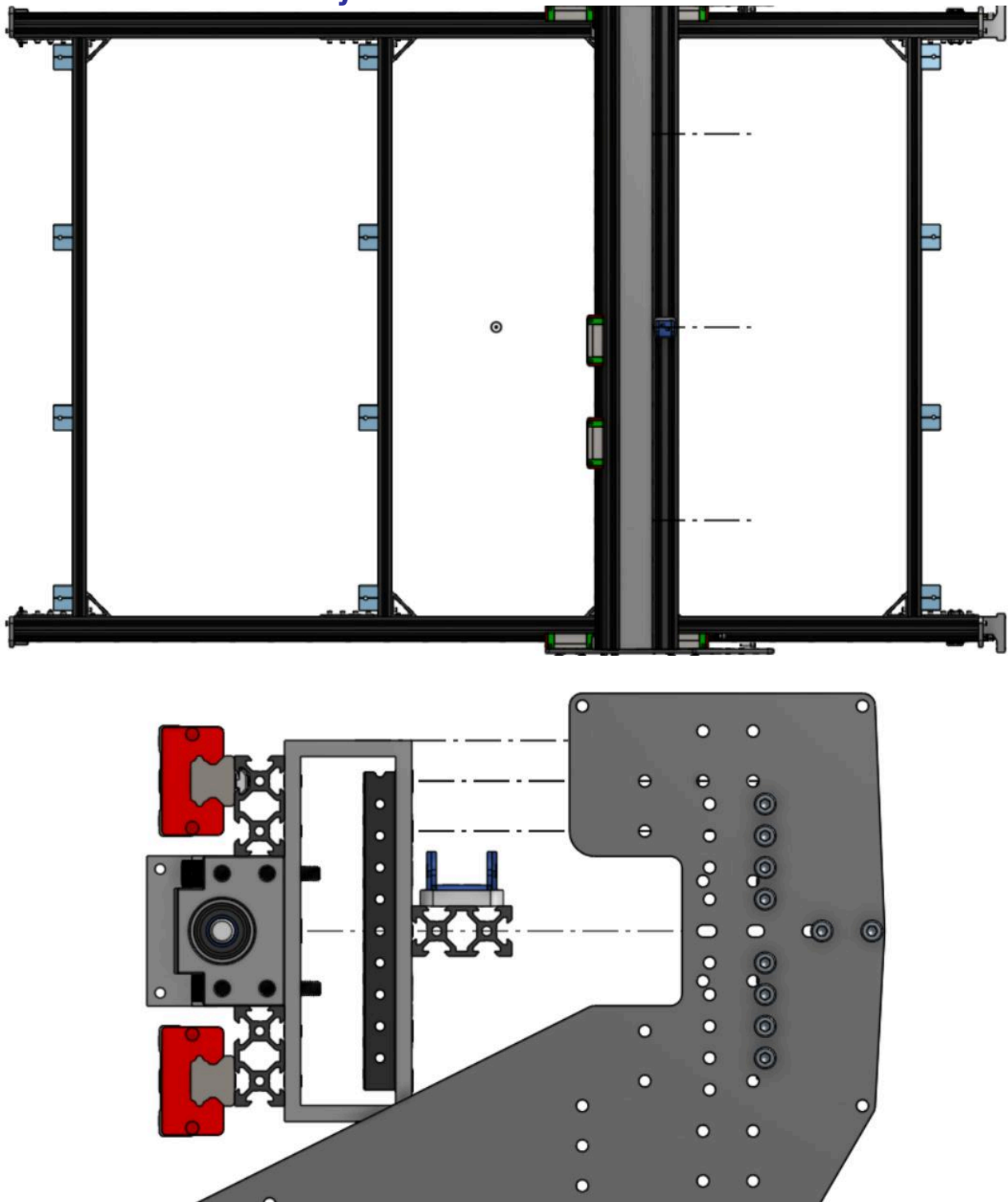


These are 10-32 because they are so easy for FRC teams to get. Don't get the screws confused with the many M5 screws used everywhere else!

This is used to attach the gantry plates more securely later.



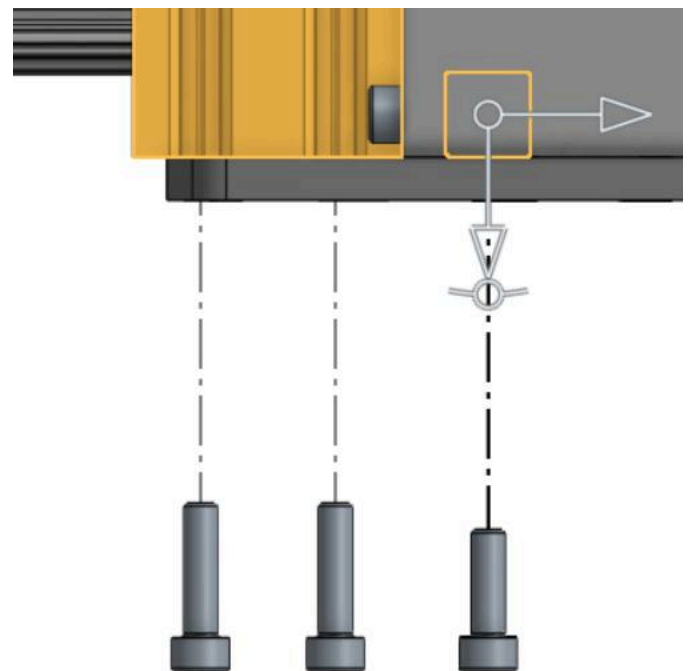
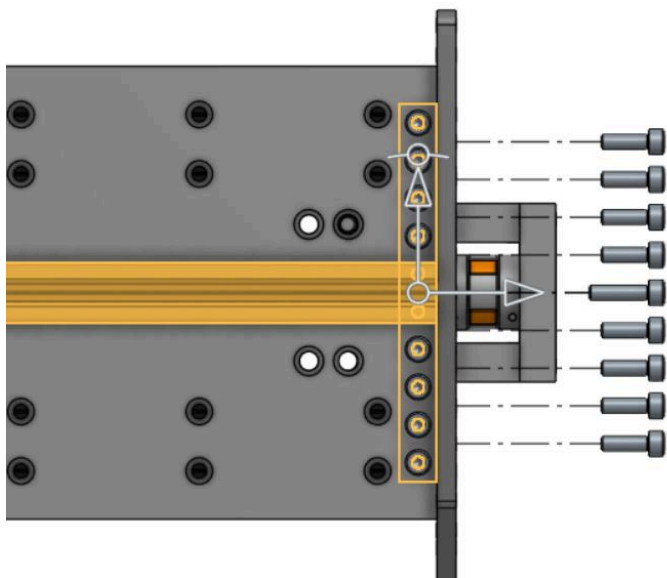
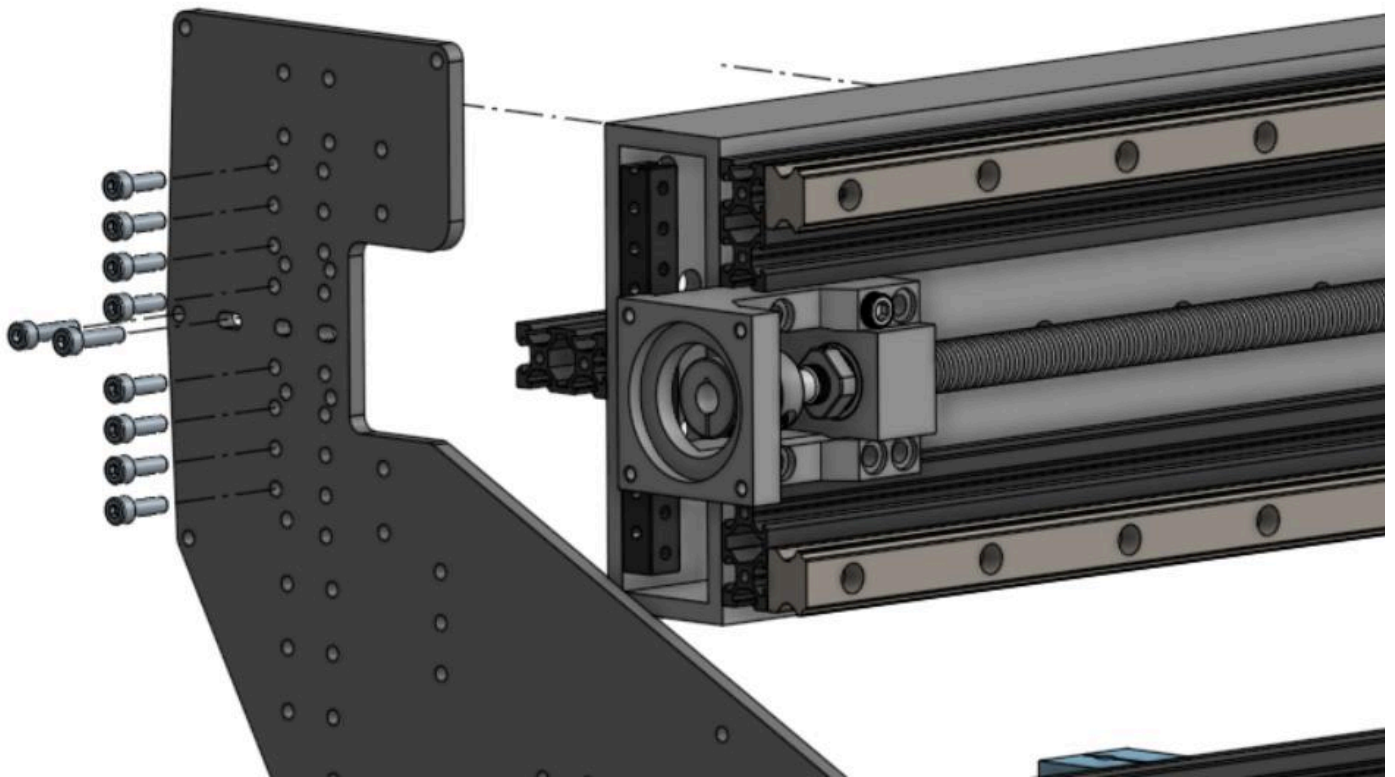
Attach the Y Axis to the Gantry Plates



Put .75" 10-32 screws into the nutstrip and 10mm M5 screws into the backbrace if you are using it.

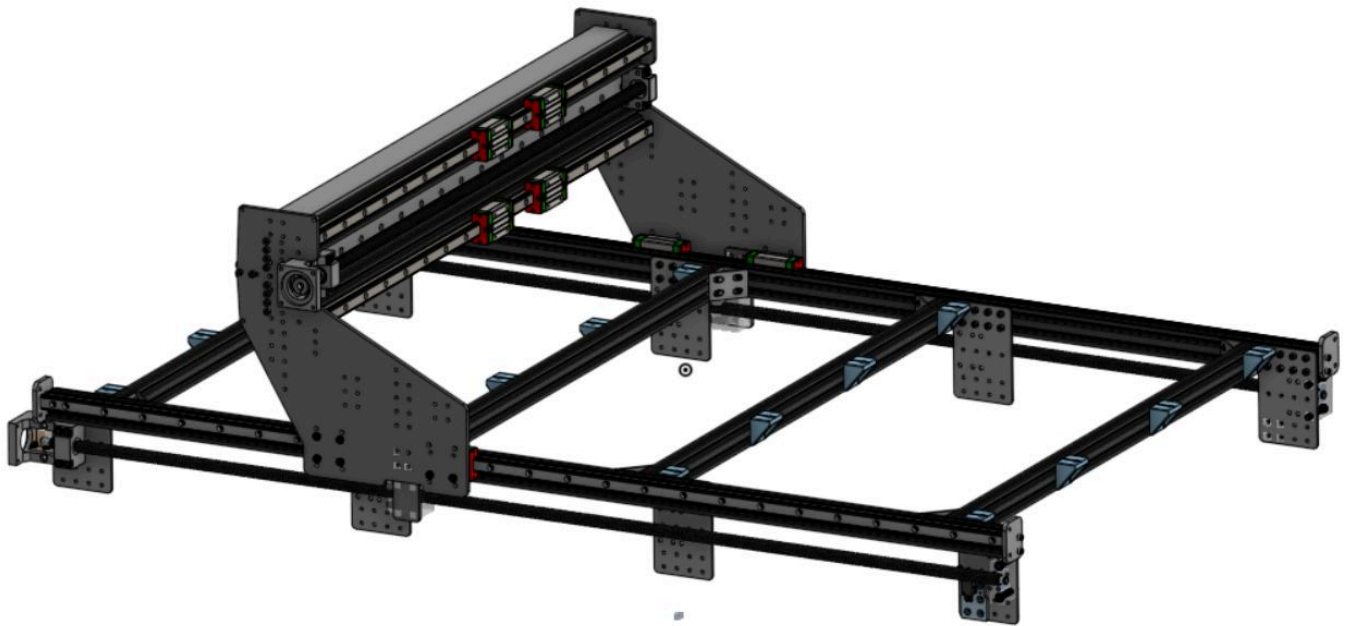


Attach the Y Axis to the Gantry Plates



End Result

It should now look like this:



After assembling the frame you are ready to install the base board (995mmx1500mm) and spoil board (also 995mmx1500mm). It's really hard to get this exactly right. It helps to have a metric tape measure and someone that's really good at cutting wood accurately. We chamfered our base board to keep the x-axis slides from dragging on it. If we had it all to do over again we would have cut a rabbet instead. You may need to do the same. Be sure the X axis moves freely before securing the base board to the frame. You can then attach the spoil board to the base board with regular wood screws. We added a 3rd level of board for extra stiffness and a simpler shape to cut/replace (30"x 48"). When you're done, don't forget to surface/level your spoil board to get accurate Z axis cuts!



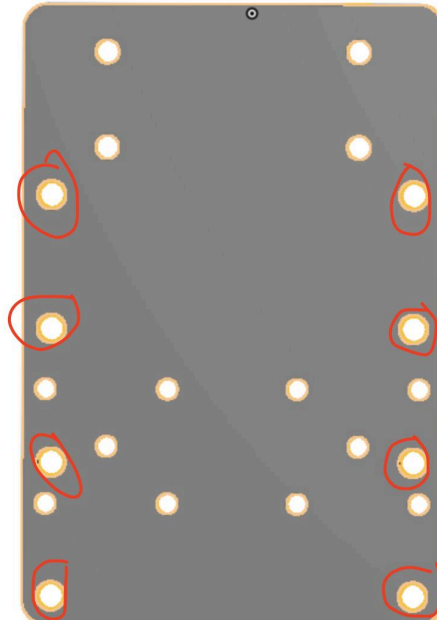
Z-AXIS

We started our FRCNC with a small 400W spindle motor and a prefabricated Z axis that used a single row of SBR16 bearing blocks. We discovered that this had more flex than we wanted so we enhanced it to add another row of bearing blocks to stiffen it, and it worked out well for us! We upgraded to a 2.2kW spindle motor so we had the added benefit of a larger mounting surface. You may want choose a Z-axis slide that already has 2 rows of bearings (recommend MGN or HGR style over the SBR style) or design your own!

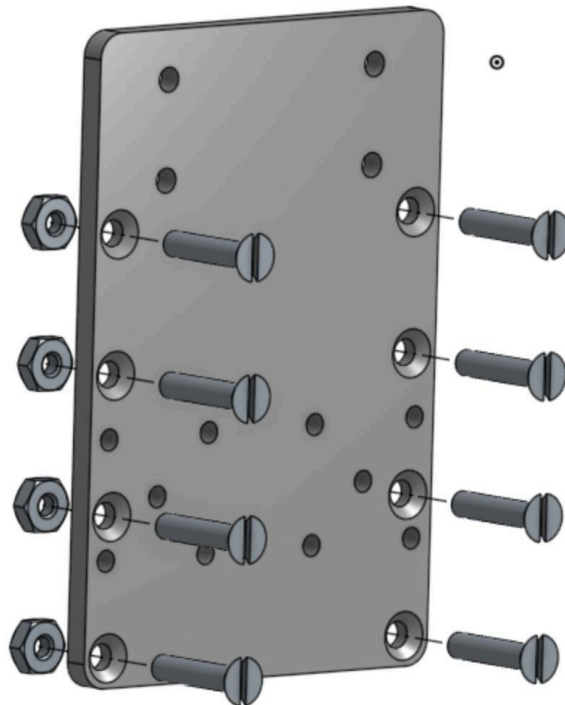


Z-AXIS

V2



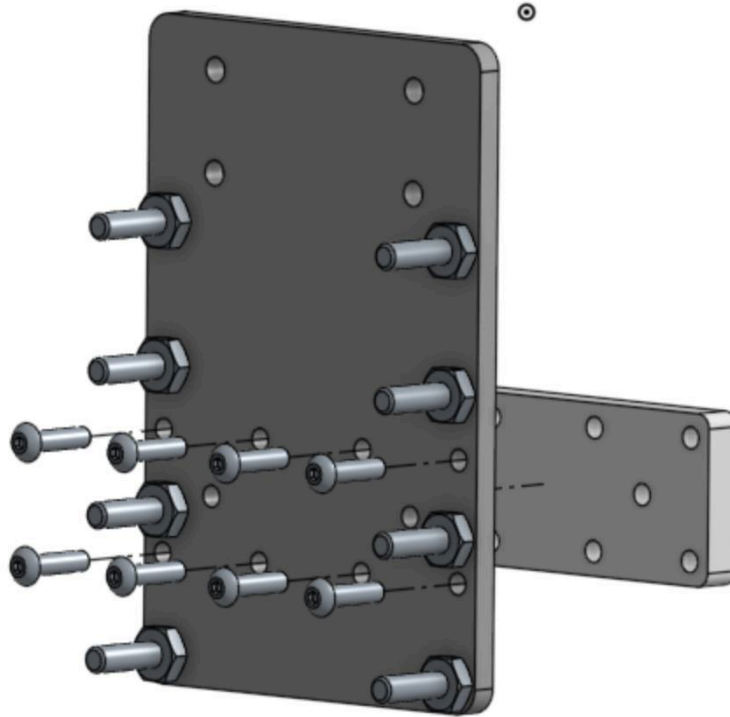
Start with this Z axis Plate. Make sure that on this side that the circled holes are countersunk.



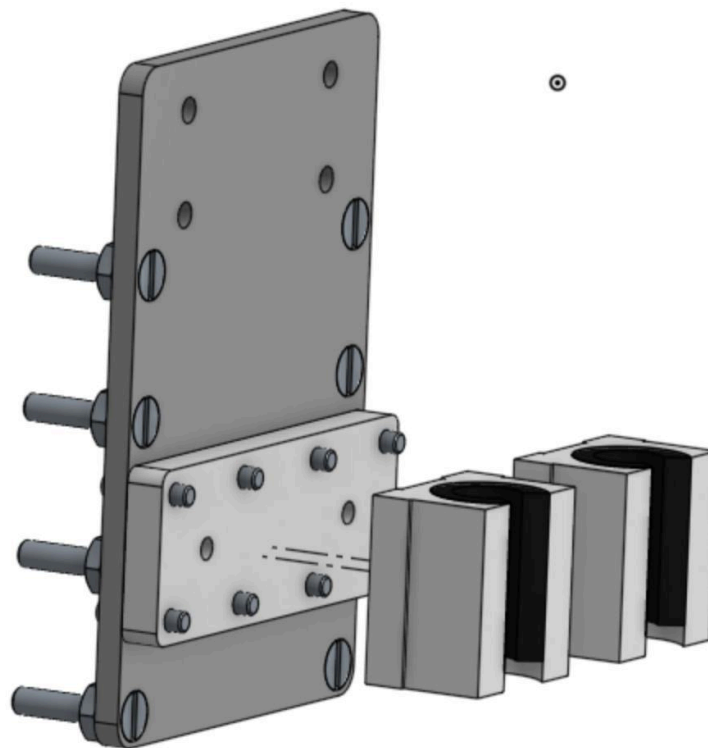
Insert eight 1/4 20 countersunk screws onto the countersunk side of the plate.
Place 1/4 20 screws on the other side and tighten, then flip the plate over



Preparing the Spindle(110 VFD Spindle Version)



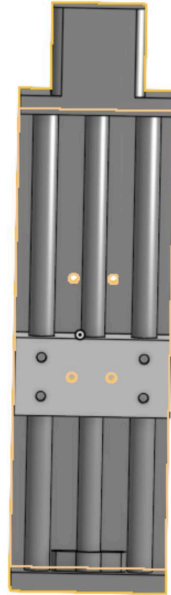
On the other side, insert eight M5 button head screws and the plate as shown.



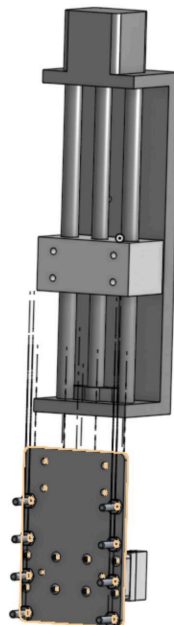
Place the bearing blocks on the screws and tighten.



Preparing the Spindle(110 VFD Spindle Version)



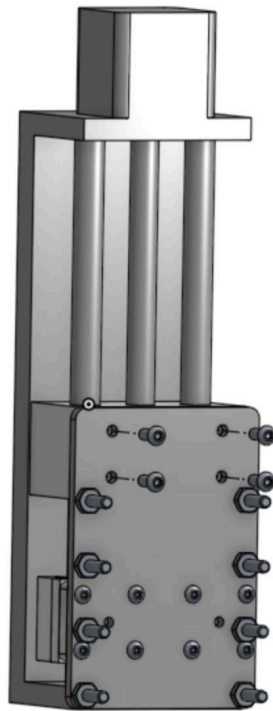
Next, get the pre-assembled Z axis and remove the bottom(the model we have is simplified, so we can't show the bottom being removed). BE SURE NOT TO REMOVE THE Z AXIS BLOCK(Light gray part).



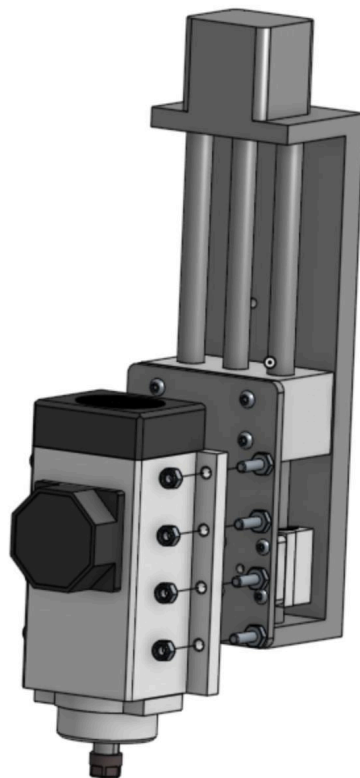
Slide the bearing blocks from your current plate onto the linear rails of the pre-assembled Z Axis.



Preparing the Spindle(110 VFD Spindle Version)



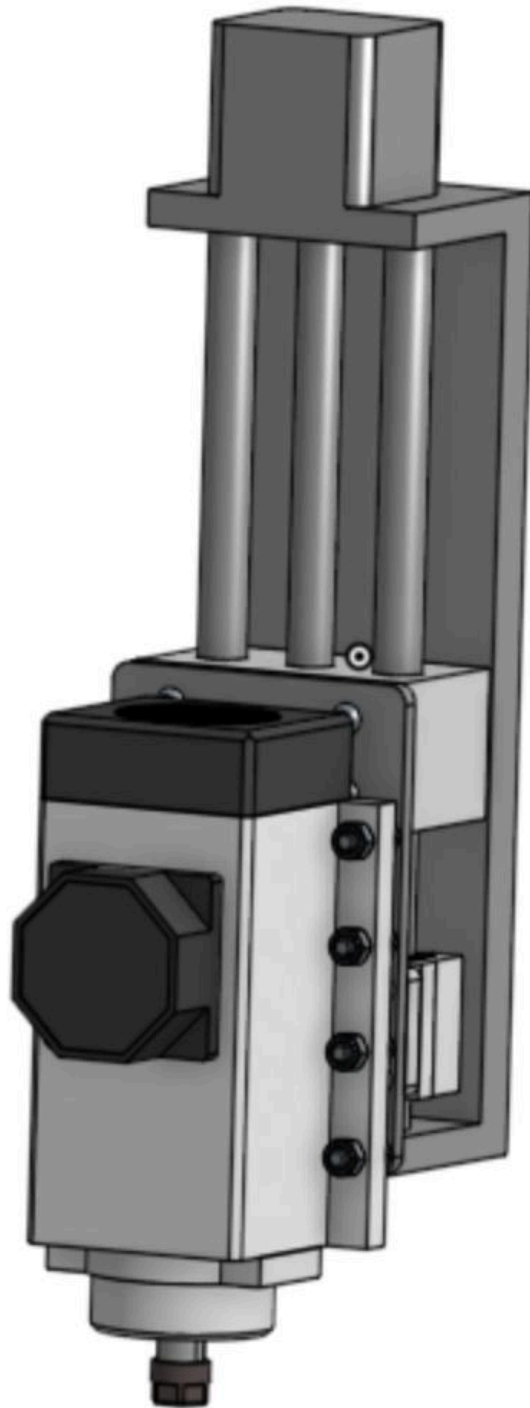
Fasten the plate to the Z axis block using four M5 screws



Add your spindle and fasten using eight ¼ 20 nylon locknuts.



Preparing the Spindle(110 VFD Spindle Version)

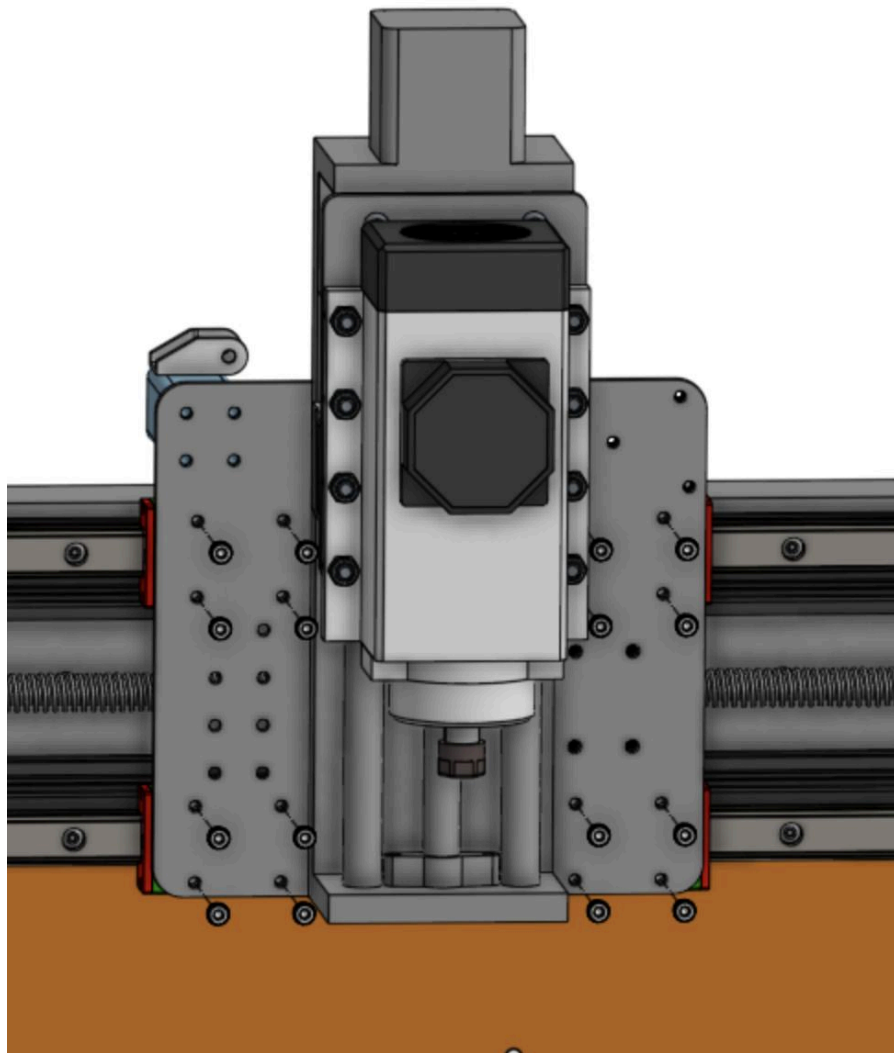


Congrats! You have now fully assembled the 110 VFD Spindle version. Skip past this next section to continue assembly.



Z-AXIS

PREPARING THE SPINDLE V1+V2

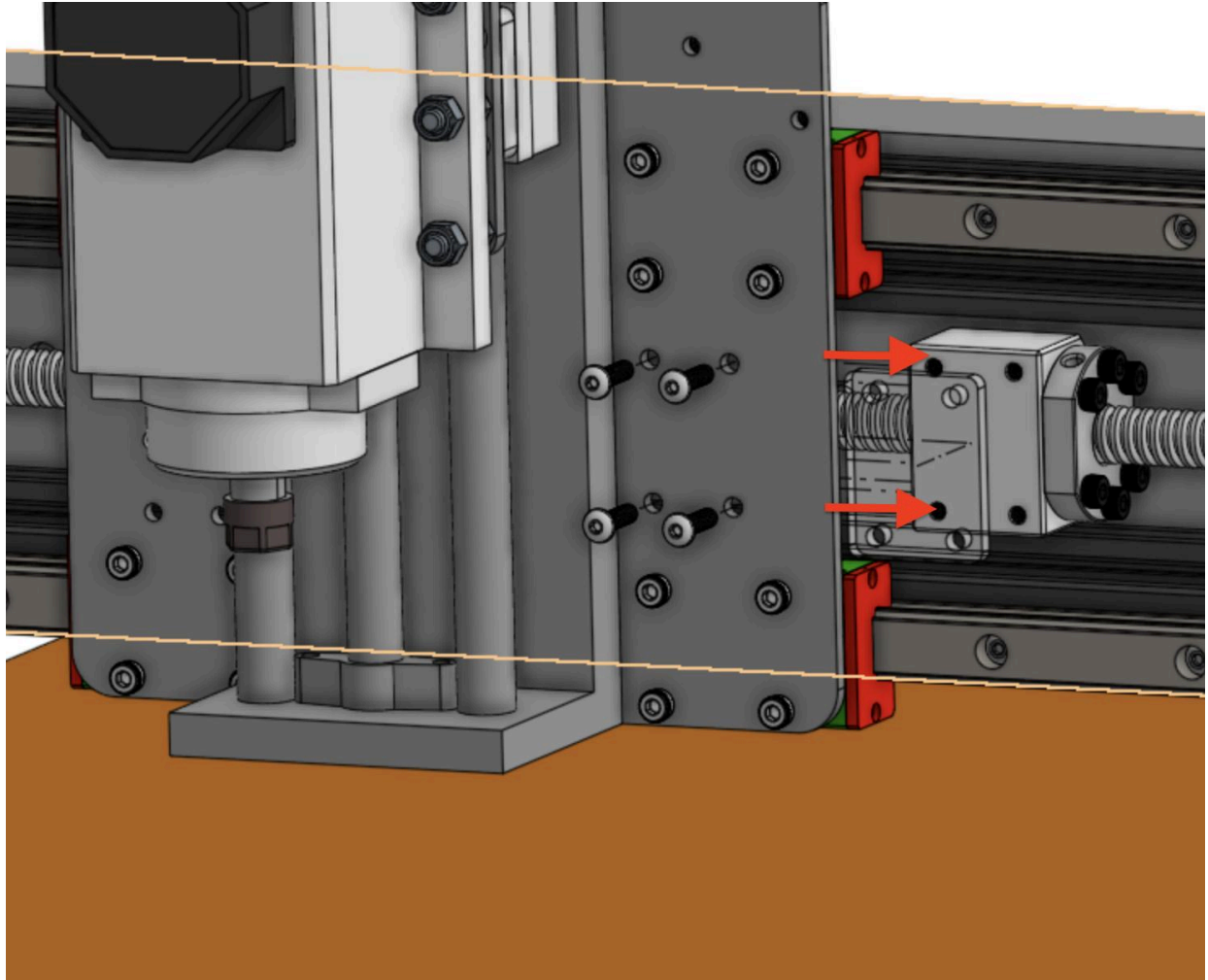


Screw your assembled spindle with the main Z-Axis plate onto the linear bearings that are on the Z Axis. Fasten with sixteen M5x10 Nuts



Z-AXIS

PREPARING THE SPINDLE V1+V2

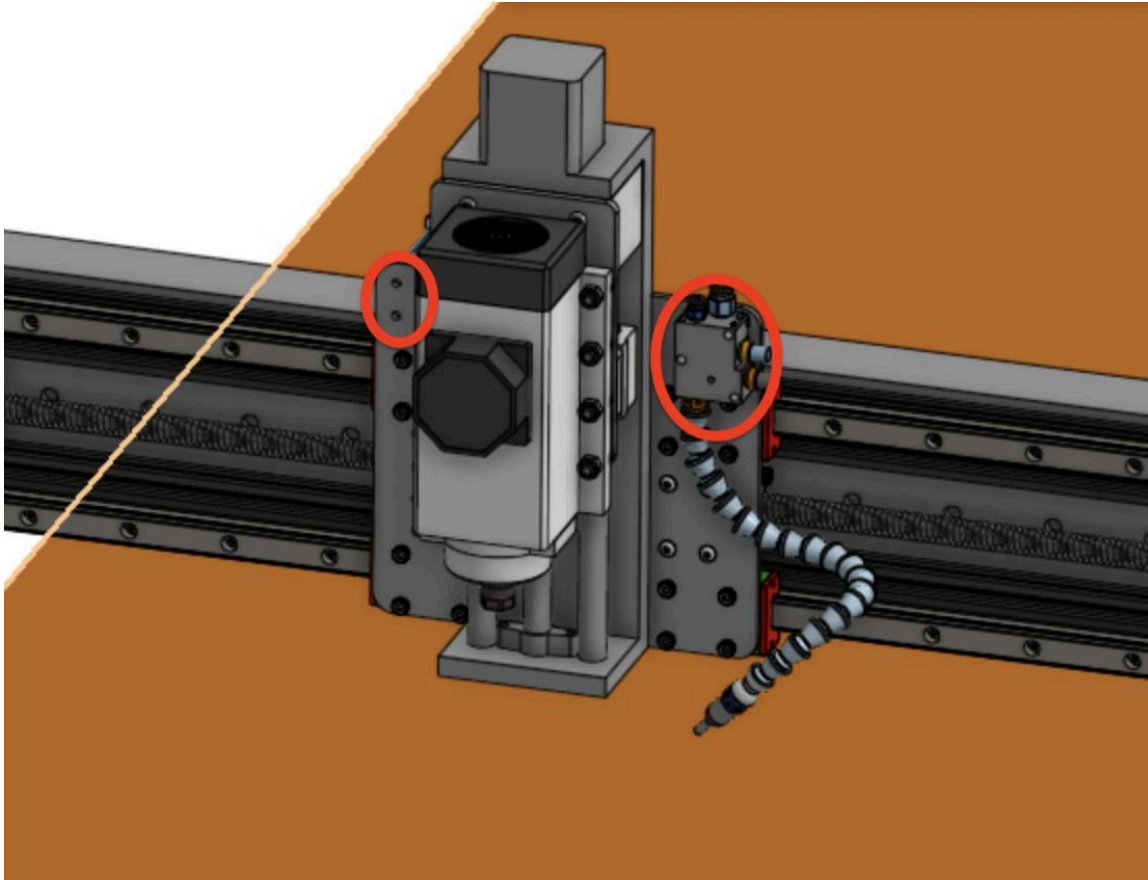


Place the spacer plate in between the main Z-Axis plate and the leadscrew. Slide the Z-Axis back over the leadscrew and make sure the spacer plate stays between. Use four M5x20 button head screws to screw the main plate to the leadscrew.



Z-AXIS

PREPARING THE SPINDLE V1+V2



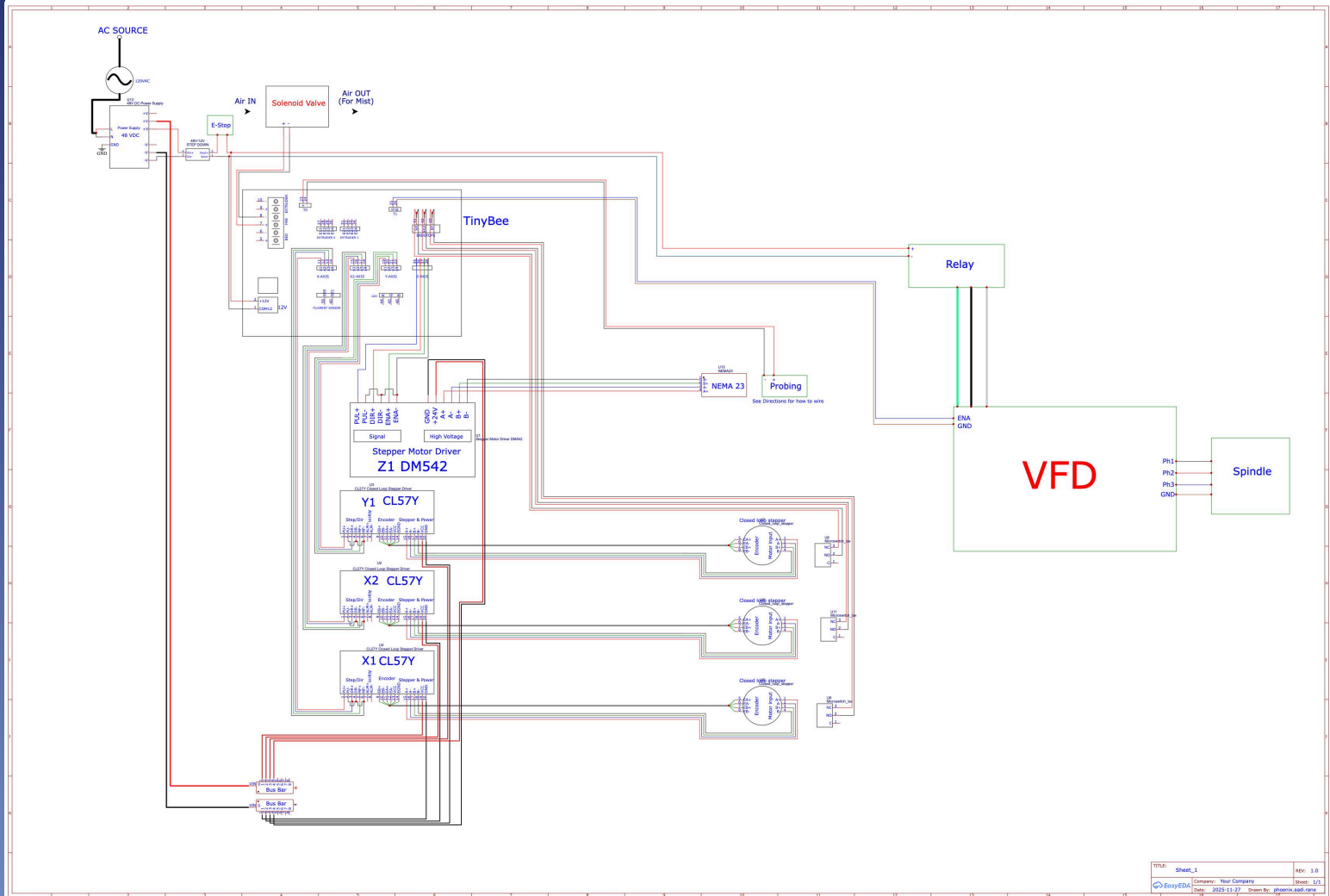
And with that you have finished the main build of the Z-Axis!

Do note a couple optional things about the Z-Axis:

- There is mounting for a mist system circled on the image
- Harder to see, on the other side is a potential mounting spot for the energy chain. Depending on whichever Y axis style you did, this could be required(Omniplate version is the only one that doesn't require this energy chain mount)



ELECTRICAL CONFIGURATION



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